Onset-Anchored Selective Computation for Resource-Constrained Vehicular Edge Networks

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ABSTRACT

Vehicular edge nodes run monitoring models continuously on hardware shared with other always-on functions. Selective computation keeps such systems within budget, but for uncertainty-driven scheduling to preserve detection coverage, the uncertainty signal must become informative around critical transitions. Continuous driving streams do not guarantee this: a classification-trained model can stay confident up to a transition, and the scheduler then suppresses the frames detection depends on. We argue that what makes uncertainty usable for scheduling is its *temporal organization* rather than calibration. We study this through the TERA ecosystem: TERA-Gen (the TERA episode Generator) generates episodes with deterministic, frame-precise onsets; AF-TOI, unlike temporal distillation, which aligns teacher predictions, shapes a compact causal model's uncertainty trajectory around the risk onset; and MHEG exploits the resulting entropy stratification at runtime. In a controlled synthetic, single-node 2×2 factorial over twelve seeds, scheduling unorganized trajectories is cheapest but does not improve coverage and worsens effective detection delay ($\mathrm{TTD}_{\mathrm{ef}}$, measured with retrospective detection markers) by 13.9% ($p = 0.001$); AF-TOI+MHEG reduces per-frame cost by 49.8% relative to Baseline Fixed (0.340 vs. 0.677 ms/frame) with $\mathrm{TTD}_{\mathrm{ef}}$ 0.099 s versus 0.570 s under Baseline+MHEG. A post-hoc calibration control shows calibration and temporal organization to be empirically dissociable under this protocol. The pipeline emits a discrete operational-risk state ℓ_t, a candidate input for vehicular edge coordination.

1. Introduction

Consider a monitoring model running on a vehicle's on-board unit. It receives a multimodal sensor stream and must flag a dangerous condition before it escalates, frame after frame, on a low-power microcontroller or automotive-grade accelerator that has no opportunity to offload the computation to remote infrastructure while the event is unfolding [1, 2]. Each frame has to be processed inside a latency budget fixed jointly by the stream's frame rate and the device's energy and thermal envelope; embedded driver-monitoring systems show that real-time inference within such a budget is achievable on automotive-grade hardware [3, 4]. Two consequences follow. First, accuracy in isolation is not the operative constraint; what matters is the ability to keep detecting critical events while the node manages its energy and compute over uninterrupted operation. Second, in vehicular edge networks – and in the ad hoc coordination architectures they may join (Section 6) – what the node emits should itself be compact and timely enough to feed downstream coordination. Driver behavior monitoring is a representative instance of this regime, and the one we use throughout. We treat the per-frame budget as given and ask a narrower question: what must be true of the model's uncertainty over time for a scheduler to spend computation selectively within that budget without putting detection at risk?

Selective computation, the runtime branch of adaptive computation [5], keeps edge inference inside its resource bounds by conditioning computational effort on an input-dependent control signal. Representative recurrent methods such as SkipRNN [6] and ACT [7] use learned gating or halting mechanisms, and video-oriented approaches such as AdaFrame [8] and LiteEval [9] use learned temporal-selection policies; these learned control variables should not be conflated with explicit predictive uncertainty. A separate family uses output confidence or predictive uncertainty directly for selective inference [10, 11], resting on the premise that low uncertainty marks a stable, redundant stretch of input that can be safely skipped. Edge deployment adds a second requirement, since resource-aware monitoring can feed alert dissemination, cooperative perception,

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and network-level policies, so the node's output has to remain useful beyond its own inference loop. Even so, the scheduler's safety turns on the premise about confidence.

That premise, usually left implicit, is that confidence is a dependable proxy for computational urgency. It is a reasonable one in activity recognition, where a model's uncertainty tends to climb as a new state begins. Continuous vehicular streaming does not guarantee it. A model trained only to separate normal from critical, with no reference to when risk begins, has no incentive to make its uncertainty behave differently in the seconds before onset than during ordinary driving. Its uncertainty trajectory can stay flat across the transition. A scheduler reading that flat signal then treats an emerging-risk frame like any redundant one and may suppress the recurrent update just as the situation turns, allocating computation away from the critical transition window, at the cost of a missed or late detection. This loss of the specific frames on which a critical event is caught, rather than a slightly higher average latency, is the operational risk we are concerned with.

Standard calibration does not detect this failure mode. Probabilistic calibration [12] checks that declared confidence matches empirical accuracy *on average*, pooled over all timesteps; it says nothing about where a given timestep sits within an episode. A model can therefore be well calibrated in this marginal sense and still be unsafe to schedule on, because marginal agreement between confidence and accuracy is silent about the ordering of uncertainty in time. That temporal ordering, near onset, is precisely what a causal scheduler consumes. Calibrating the output probabilities cannot supply information the trajectory never encoded.

We call the property that a scheduler actually needs *temporally conditioned operational reliability*: the ability of the uncertainty signal to distinguish transition regions from stable ones at the moment the scheduler acts. Neither standard calibration nor existing analyses of non-stationary entropy dynamics [13, 14] treat this as a prerequisite for scheduling. It is also the property that would let a node summarize its situation for higher-level network functions without shipping raw frames or full predictive trajectories.

The causal constraint sharpens the problem. Onboard inference is strictly causal: a decision at time t may use only the history up to t, so any failure to separate a transition from a preceding stable interval shows up immediately as an unsafe suppression, with no future frame to recover from it. The episodes are also temporally heterogeneous, since the same stream mixes gradual, multi-second build-ups with abrupt onsets under 600 ms [1, 15]. On naturalistic data these differences are hard to see at all, because an onset annotated to within $\approx$0.31 s at 9.6 fps blurs scheduler behavior below that scale, especially when a missed episode carries no explicit penalty [2, 15, 16].

Our hypothesis (Figure 1) is that this cluster of difficulties is addressed less by tuning the scheduler's thresholds than by deliberately shaping the model's predictive trajectories during offline training so that they differentiate in time. Once the trajectories carry that structure, the model's own uncertainty becomes informative about the proximity of a critical event, and the runtime allocator receives an uncertainty signal with operational meaning. On this reading, the apparent conflict between efficiency and coverage is relieved by reorganizing the model's internal dynamics rather than by threshold engineering. The same organized signal also lets the node report a discrete operational state instead of a bare, local confidence value.

By an *informative scheduling signal* we mean an uncertainty trajectory whose level separates episode phases: low and stable during redundant intervals, rising ahead of and during risk transitions. This is the property depicted in the central block of Figure 1: without it, the entropic component cannot distinguish transition regions from stable intervals, and only the unconditional kinematic trigger remains protective; with it, thresholding entropy directs computation to the frames surrounding risk onset.

The paper contributes three components that together isolate and exploit this property. *TERA-Gen* (the *TERA episode Generator*, the synthetic-data component of the ecosystem) generates synthetic episodes with a deterministic onset calibrated against SHRP2 odds ratios [15]; by fixing the onset instant it removes the annotation variability that would otherwise make frame-precise induction and evaluation impossible. *AF-TOI* is an onset-oriented induction protocol that shapes the causal model's predictive dynamics so they separate stable regions from transition windows, giving the scheduler an uncertainty signal that means something in time. Its training target is the temporal shape of the uncertainty trajectory around t_0, not agreement with the reference model's predictions (Section 2.6). *MHEG* is the runtime scheduler that consumes that signal; it pairs an unconditional kinematic trigger, which forces computation on abrupt physical change, with an entropic gate whose usefulness depends on the induced organization, and whose behavior when that organization is *absent* is measured directly through a negative control rather than assumed. Beyond choosing between a full recurrent update and state reuse, MHEG emits a discrete operational-risk state $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$ that is low-bandwidth, interpretable, and produced causally. The three components constitute the *TERA* (*Temporal Evaluation and Risk Assessment*) ecosystem, in which organization is induced offline and allocation happens online on the device.

The controlled experiments provide evidence consistent with this hypothesis under the evaluated synthetic conditions, and the most telling result is the negative one. On unorganized trajectories, adding the scheduler does not improve coverage (FR 0.033 → 0.035) and worsens TTD_{ef} by 13.9%: the scheduler remains active, but its entropic component cannot discriminate transition windows from stable intervals. Give the same scheduler AF-TOI-organized trajectories and per-frame inference cost falls by 49.8% relative to Baseline Fixed (0.340 vs. 0.677 ms/frame), while TTD_{ef} is 0.099 s against 0.570 s under Baseline+MHEG. A post-hoc calibration control then addresses the obvious alternative explanation: under this protocol, rescaling probabilities improves calibration without reorganizing entropy in time, so calibration alone does not account for the observed temporal organization and the two properties are empirically dissociable under this protocol. MHEG thus works both as a local selective-computation mechanism and as a source of a compact operational state that could later connect embedded inference to resource-aware coordination.

The remainder of this paper is organized as follows. Section 2 reviews related work and positions our contribution. Section 3 presents the TERA framework: deterministic temporal anchoring (TERA-Gen), temporal organization induction (AF-TOI), and the runtime scheduler (MHEG). Section 4 reports the experimental evaluation, including the factorial study, the robustness controls, and the post-hoc calibration control. Section 5 discusses implications and limitations, Section 6 outlines architectural implications for vehicular edge computing, and Section 7 concludes.

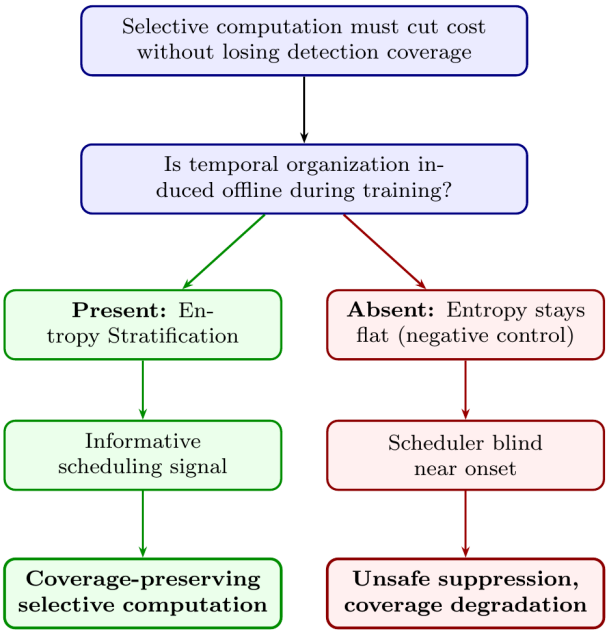


Figure 1: Conceptual hypothesis investigated by TERA. Offline temporal organization induces entropy stratification across episode phases, turning predictive uncertainty into an informative scheduling signal. Without this organization, entropy remains weakly differentiated and selective computation may suppress inference near risk onset. The "informative scheduling signal" block denotes phase-stratified entropy: low in stable intervals and elevated near onset, which is the property the runtime gate thresholds.

2. Related Work

2.1. Resource-Constrained Edge AI in Vehicular Edge Computing

Vehicular Edge Computing (VEC) and Mobile Edge Computing (MEC) relocate inference from the cloud to on-board or roadside units, trading round-trip latency for tight local budgets in energy, memory, and bandwidth [1, 2, 17]. The motivation is concrete: on embedded automotive platforms such as the STM32H7, Cortex-M55, and SA8155P, the monitoring model typically shares the processor and the energy budget with other always-on vehicle functions, and production-scale perception stacks are substantially larger than the deliberately compact model studied here, so some form of runtime allocation is unavoidable. The deployed model in this paper is kept small (≈58 K parameters) precisely to isolate the allocation mechanism; the absolute savings it yields grow with the size of the predictor the mechanism gates and with the degree of processor sharing.

Edge-native ITS work has answered mostly on the model side – quantization [3] and resource-aware architecture [18] – while the broader VEC and MEC literature works at the network side, on static compression or task placement across nodes. Both leave

a complementary question comparatively untouched in safety-critical settings: when, within a single continuous on-device stream, is it worth spending a full inference step (Section 2.4)? That single-node question is the one we take up first, because it has to be answered before uncertainty can be relied upon as a coordination signal.

A clarification on scope is warranted: vehicular networks (e.g., IEEE 802.11p/DSRC, C-V2X) and general-purpose mobile ad hoc networks differ in protocol stack, mobility model, and infrastructure assumptions, and we do not treat them as interchangeable. TERA operates at the computation layer of a single node – deciding when to spend an inference step – and is agnostic to the underlying communication protocol; the operational-risk state ℓ_t it emits is a compact payload for dissemination by the underlying network. Protocol-level concerns such as medium access, routing, and dissemination latency are outside the scope of this paper.

2.2. Trustworthy and Validated Edge AI

Efficiency alone is insufficient at the edge; dependability is an equally important requirement. Throughout this paper, "trustworthy" is used in a restricted, operational sense: a selective-computation policy is trustworthy when it preserves critical-episode coverage in every evaluated seed (FR $\leq$ 0.05 per seed, not merely on average) while suppressing computation, under the controlled conditions evaluated here. Because an aggregate failure rate can conceal unsafe behavior in individual replications, the coverage constraint is evaluated at the seed level rather than only on the pooled mean. Under this criterion, Baseline+MHEG satisfies the aggregate budget (FR = 0.035 on average) but violates the bound in three of the twelve seeds (up to 0.158), whereas AF-TOI+MHEG remains within it in all twelve (max 0.017). We make no claim regarding broader dimensions of trustworthiness (security, privacy, adversarial robustness), which are out of scope (Section 5).

Wang et al. [19] argue that resource limits, heterogeneous environments, and diverse requirements jointly determine whether an edge system can be trusted, and ML-safety work on uncertainty quantification for autonomous vehicles makes the same point from the modeling side [20–22].

The recurring lesson is that dependability has to be demonstrated under deployment conditions, not read off benchmark accuracy. A scheduler that suppresses inference is squarely in this reliability-critical category, since its skip decisions decide whether a safety-relevant event is still caught once computation is cut. What validating such a scheduler requires, concretely, is evidence that the uncertainty signal it reads actually separates transition regions from stable ones – and a direct way to establish the role of this property is to remove it experimentally, which is the function of our negative control. To the best of our knowledge, dependability at this level, the internal structure of the trajectory the scheduler consumes, has not been examined as a validation target in the trustworthy edge AI literature.

2.3. Temporal Risk Detection in Continuous Driver Monitoring

Automated driver behavior monitoring in naturalistic environments is well established. The DMD dataset [2] provides dense annotations for seven behavior classes with front-camera and physiological fusion, and DADA [16] extends coverage to eye tracking in pre-collision scenarios. Both, however, define onset by retrospective human annotation with variability of $\approx$0.31 s at 9.6 fps.

That imprecision directly affects edge resource-allocation design. Undetected episodes vanish from latency averages without penalty, and temporal granularity is too coarse to separate transition zones from stable periods. Without a deterministically anchored onset, one cannot tell whether the scheduler invested resources at emergence windows rather than redundant intervals, which is the central decision the edge system must make.

The SHRP2 study [15] documents temporally heterogeneous risk trajectories (multi-second gradual transitions down to sub-0.6 s abrupt onsets) with up to twelvefold collision risk [1]; dominant validation protocols rely on aggregate recognition metrics and evaluate neither onset timing nor scheduler cost differentiation.

A more recent group of systems settles the feasibility question on the hardware we care about. Salakapuri et al. [23] run an integrated distraction and road-object detector on an automotive-grade platform at real-time rates, and lightweight time-series and TinyML models report the same under tight budgets [4, 24]; our own Safe-Drive system pairs behavior detection with seatbelt segmentation for per-frame monitoring [25]. On-device viability has therefore been empirically demonstrated. What these evaluations report, though, is aggregate recognition accuracy and latency – not when a risk is detected relative to its onset, and not what it costs to suppress computation during the transition that leads up to a critical event. The metrics required to expose allocation errors at the scheduler level are therefore not reported. The combination that would expose them – a deterministic onset, an explicit penalty for a missed episode, and suppression treated as a resource-allocation choice – is what TERA-Gen and $\mathrm{TTD}_{\mathrm{ef}}$ are designed to supply.

2.4. Adaptive Inference and Selective Computation in Embedded Systems

Adaptive computation families such as dynamic networks [5, 26], conditional computation and early exiting [11, 27–29], and split computing [30, 31] cut

inference cost by adapting to the input. Two strategies dominate: structural compression (quantization, pruning [3]) and runtime scheduling of computation, which spans learned adaptive computation with gating or halting mechanisms (SkipRNN [6], ACT [7]), learned temporal-selection policies (AdaFrame [8], LiteEval [9]), and confidence- or uncertainty-driven selective inference (selective classification [10]); the control variables of the first two families are learned signals rather than explicit predictive uncertainty. Kaya et al. [32] show that overthinking, where computation continues past correct classification, stems from reliance on instantaneous confidence rather than trajectory structure. The present work sits in the on-device branch, where the decision is not *where* to run inference but *when* to invest it within a continuous stream.

These efficiency gains are real, but they rest on the assumption that high confidence signals absent risk and therefore safe suppression. Recent surveys confirm that output confidence and entropy remain the dominant exit or skip criteria [11]. The assumption holds in activity recognition, where predictive trajectories exhibit uncertainty sensitive to state transitions, giving the scheduler a reliable allocation signal.

In the vehicular-monitoring setting studied here, this assumption can break down. A model may stay confident just before onset not because the situation is stable but because its internal trajectories lack the temporal differentiation that would distinguish transition zones from preceding stable periods. A scheduler designed for activity recognition may then invest in redundant timesteps while suppressing inference at emergence windows, producing an unsafely optimistic efficiency gain. Current design criteria evaluate scheduling through throughput, latency, and deployment metrics without treating temporal detection coverage as an explicit safety constraint. Whether entropy-driven suppression is safe when a missed critical episode carries a real cost is therefore left open, and it is exactly the case AF-TOI and MHEG are built to address.

2.5. Uncertainty Calibration and Operational Reliability

Trustworthy selective computation needs uncertainty that differentiates transition regions, not just uncertainty that matches aggregate frequencies. Predictive uncertainty estimation is established [33–35], and probabilistic calibration, the alignment between declared confidence and empirical accuracy [12], is an accepted automotive-safety criterion [36, 37], with Minderer et al. [38] showing it does not improve monotonically with architecture. Calibration, though, is marginal: a perfectly calibrated model may still schedule computation unsafely if trajectories do not differentiate uncertainty at transitions. Braverman et al. [13] and TRUST [14] confirm the non-stationary character of entropy dynamics without formalizing near-onset behavior as a scheduling signal.

A related question, which the reviewed literature does not seem to have addressed directly, is whether improving global calibration also changes how uncertainty is organized in time. There is a structural reason to expect it does not. Post-hoc techniques – temperature scaling [12], Platt scaling [39], isotonic calibration [40] – rescale the output probabilities without touching the weights, and therefore without touching the internal trajectory that a causal scheduler reads. A monotonic rescaling preserves the ordering of the probabilities, but not, in general, the ordering of the binary entropies, since $H(p)$ rises up to $p = 0.5$ and falls beyond it; whether a given calibrator reorganizes the temporal entropy profile is therefore an empirical question rather than a mathematical impossibility. We are not aware of a study in the adaptive causal scheduling literature that tests this divergence experimentally, and we do so here as an explicit control.

2.6. Temporal Structure Transfer and Sequential Modeling

Temporal structure transfer [41–43] and online sequential architectures [44, 45] advance efficiency and accuracy, but none asks whether the transferred structure enables trustworthy runtime allocation. Conventional transfer objectives spread the alignment signal uniformly, pushing compact models to reproduce average predictive behavior rather than uncertainty dynamics that separate stable from transition regions, so entropy stratification for trustworthy selective computation stays unaddressed. Prior works also target segmented offline clips where the full sequence is available; causal streaming invalidates sequence-level normalization and global attention, so organizational gains from those contexts do not carry over automatically. Explicit induction mechanisms are required.

The methodological distinction of AF-TOI relative to these families can be stated explicitly. Temporal knowledge distillation and sequence-level transfer optimize agreement with a teacher's *predictions*, spreading the alignment signal uniformly over time; their objective is accuracy or compression. AF-TOI instead optimizes the *shape of the uncertainty trajectory* around a deterministic onset: the teacher contributes only a probability-space alignment term (Eq. 7), while onset-anchored shaping penalties, concentrated by the exponential focus of Eq. 8 anchored to t_0, act on the student's own trajectory. The training target is entropy stratification across episode phases, not fidelity to teacher outputs. Uncertainty-aware inference methods consume uncertainty at runtime but do not reshape its temporal structure during training, and adaptive/selective computation methods assume an informative confidence signal rather than inducing one. To the best of our knowledge, using onset-anchored

structural alignment to *create* the scheduling signal that a selective-computation gate later consumes is the specific contribution of AF-TOI.

2.7. Gaps and Positioning

Read across these bodies of work, the same gap recurs, and Table 1 localizes it in the **Temp. Org.** and ΔH^* columns. Driver monitoring established the operational stakes but provides neither a deterministic onset nor an explicit miss penalty, so the cost of suppressing computation at a transition window cannot be measured. The adaptive inference literature delivered the efficiency but inherited the premise that high confidence means absent risk – a premise that fails once the internal trajectory is temporally flat. Studies of temporal entropy dynamics [13, 14] showed that entropy is non-stationary but stopped short of treating onset dynamics as a scheduling property. Temporal structure transfer built compact models without asking whether the transferred structure was the kind a scheduler could safely allocate on. In the table, the ΔH^* column is ✗ for every prior entry and ✓ only for TERA, and **Temp. Org.** is uniformly ✗ or ○ across prior work. On the evidence we surveyed, we are not aware of prior work that uses temporal entropy dynamics near a deterministic onset as a scheduling signal under an explicit miss-coverage constraint, or that treats temporal organization as a structural precondition along the chain *Entropy Stratification* → *Informative Scheduling Signal* → *Coverage-Preserving Selective Computation*.

These gaps form a dependency chain. Onset dynamics cannot be characterized without deterministic annotation; entropy does not gain discriminative power at transitions without onset-aware training; and, in the setting evaluated here, an entropy-driven gate cannot allocate computation reliably around the onset when the entropy trajectory itself is not informative. This does not imply that every selective-computation mechanism requires temporally organized entropy: periodic-refresh and learned-gating baselines provide alternative allocation mechanisms, examined in Section 4.11. TERA treats temporal organization as the structural precondition for entropy-driven selective scheduling under the explicit coverage constraint evaluated here: AF-TOI promotes the trajectory structure required for entropy stratification, and MHEG exploits the resulting signal to allocate computation selectively and safely at the edge.

The columns of Table 1 are read as follows. **Onset**: deterministic onset annotation for training and evaluation; **TTD**: detection-latency evaluation relative to event start; **Cov.**: critical-episode coverage with an explicit miss penalty; **Causal**: strictly causal online inference; **Temp. Org.**: temporal organization treated as a first-class analytical object (its induction, analysis, validation, and exploitation); ΔH^*: near-onset entropy dynamics used as a scheduling signal under coverage constraints; **Eff.**: efficiency analysis; **Edge**: embedded or vehicular-edge deployment; **Real**: evaluation on naturalistic data or physical hardware. A ✓ denotes a property that is present, ○ one that is partial or implicit, and ✗ one that is absent.

On the use of synthetic data in Table 1 The Real Eval. column marks this work with ✗, an explicit methodological choice rather than an oversight. Controlled evaluation requires TERA-Gen, whose deterministic onset annotation removes the ≈0.31 s variability of naturalistic datasets such as SHRP2, DMD, and DADA [2, 15, 16]; at that variability, scheduler differences below 0.3 s would be indistinguishable from annotation noise. Calibrating TERA-Gen against SHRP2 odds ratios preserves the statistical structure of real risk trajectories while removing the confounder that would otherwise block frame-precise attribution of scheduling gains to temporal organization. Embedded driver-monitoring systems demonstrate that real-time inference on automotive-grade hardware is feasible under tight per-frame budgets [3, 4], so the operational setting the scheduler targets is hardware-grounded even though the temporal-organization effect is measured in a controlled synthetic environment. Transfer to naturalistic driving data is the primary direction for future validation (Section 5).

Table 1
Comparison between the TERA ecosystem and representative related work. Column definitions are given in the paragraph preceding the table; marks are ✓ (present), ○ (partial or implicit), and ✗ (absent).

Work	Onset	TTD	Cov.	Causal	Temp. Org.	ΔH^*	Eff.	Edge	Real
Driver monitoring and risk detection									
Ortega et al. (DMD) [2]	✗	✗	✗	✗	✗	✗	✗	✗	✓
Bouhsissin et al. [1]	✗	✗	✗	✗	✗	✗	✗	✗	✓
Salakapuri et al. [23]	✗	✗	✗	✗	✗	✗	✓	✓	✓
Driver state TinyDL [4]	✗	✗	✗	✗	✗	✗	✓	✓	✓
Cognitive distraction [24]	✗	✗	✗	○	✗	✗	✓	✓	✓
Adaptive inference and selective computation									
Hariharan et al. [3]	✗	✗	✗	○	✗	✗	✓	✓	✓
Campos et al. (SkipRNN) [6]	✗	✗	✗	✓	✗	✗	✓	✗	✓
Wu et al. (AdaFrame) [8]	✗	✗	✗	✗	✗	✗	✓	✗	✓
Graves (ACT) [7]	✗	✗	✗	○	✗	✗	✓	✗	✓
Wu et al. (LiteEval) [9]	✗	✗	✗	✗	✗	✗	✓	✗	✓
Temporal entropy dynamics and uncertainty calibration									
Braverman et al. [13]	✗	✗	✗	✗	○	✗	✗	✗	✓
Dera et al. (TRUST) [14]	✗	✗	✗	○	○	✗	✗	✗	✓
Temporal structure transfer and sequential modeling									
Zhao & Krähenbühl [44]	✗	✗	✗	✓	○	✗	✗	✗	✓
Xu et al. (LSTR) [45]	✗	✗	✗	✓	○	✗	✗	✗	✓
Zheng et al. [43]	✗	✗	✗	✗	○	✗	○†	✗	✓
Bajestani & Yang [42]	✗	✗	✗	✗	○	✗	✗	✗	✓
This work (TERA)	✓	✓	✓	✓	✓	✓	✓	○†	✗

† Partial marks: partial efficiency evaluation only (prior work); for TERA, embedded deployment is characterized analytically and on a reference GPU (Section 4.13), not on physical embedded hardware. TERA is evaluated on synthetic data (TERA-Gen) and a reference GPU; naturalistic and on-device validation remain future work (Section 5). Each work is cited in the corresponding subsection of Section 2.

3. From Temporal Organization to Onset-Anchored Selective Computation

Embedded risk detection processes signals sequentially without access to future observations, and this causal constraint governs TERA. The on-board model operates strictly causally at deployment, while coverage stability comes from the temporal organization induced offline and then exported to the device.

The pipeline ingests a continuous stream $\mathbf{x} = (x_1, x_2, \ldots)$, where each $x_t \in \mathbb{R}^D$ encodes visual, kinematic, contextual, and acoustic observations. The output y_t depends only on the history $\{x_1, \ldots, x_t\}$, which distinguishes the embedded model from bidirectional architectures. The predictive core is a two-layer unidirectional LSTM with recurrent-update cost $\mathcal{O}(4(DH+H^2) + 8H^2)$ for the two stacked layers (≈57.2 K MACs per executed timestep for D=31, H=64; Section 3.5), compatible with automotive-grade edge budgets, where H denotes the recurrent hidden width, distinct from the entropy $H(\cdot)$ used by the scheduler.

The sensor stream at the input of Figure 2 is the frame-synchronous concatenation of the four on-board modality groups detailed in Section 3.1: behavioral-visual channels from the cabin-facing detector, kinematic channels (speed, longitudinal accelerations, gyroscope), driving-context channels (road type, precipitation, illumination, traffic density), and acoustic event channels, plus two detector-reliability indicators – yielding $x_t \in \mathbb{R}^{31}$ per frame at ≈9.6 fps.

The claim that trajectory organization is what lets MHEG schedule safely is tested, not assumed: it is assessed through entropic separation, coverage stability, allocator selectivity, and $\mathrm{TTD}_{\mathrm{ef}}$ across the factorial configurations. AF-TOI supplies the organization offline; the negative control removes it, so its role shows up as the difference between the two.

Figure 2 summarizes the operational architecture. The framework separates offline induction from online inference. Offline, TERA-Gen, the non-causal reference model, AF-TOI, entropy stratification, and threshold calibration produce an organized causal model and calibrated thresholds. Online, the deployed causal model and MHEG use this artifact to perform selective computation under coverage-preservation constraints.

Operational problem formulation Table 2 lists six design objectives, of which one arbitrates the central conflict: $\mathrm{TTD}_{\mathrm{ef}}$. Traditional latency metrics such as $\mathrm{TTD}_{\mathrm{det}}$ are computed only over detected episodes and ignore complete misses, which can make a system that misses a fraction of events report a low average latency on the rest. A scheduler may thus reduce apparent cost without changing the miss rate, and a coverage-blind metric would not notice. The effective detection delay integrates timeliness and coverage,

$$\mathrm{TTD}_{\mathrm{ef}} = \mathrm{TTD}_{\mathrm{det}} + \mathrm{FR} \times T_{\mathrm{ep}} \tag{1}$$

where $\mathrm{TTD}_{\mathrm{det}}$ is the mean latency over detected episodes, FR is the missed-episode proportion, and T_{ep}

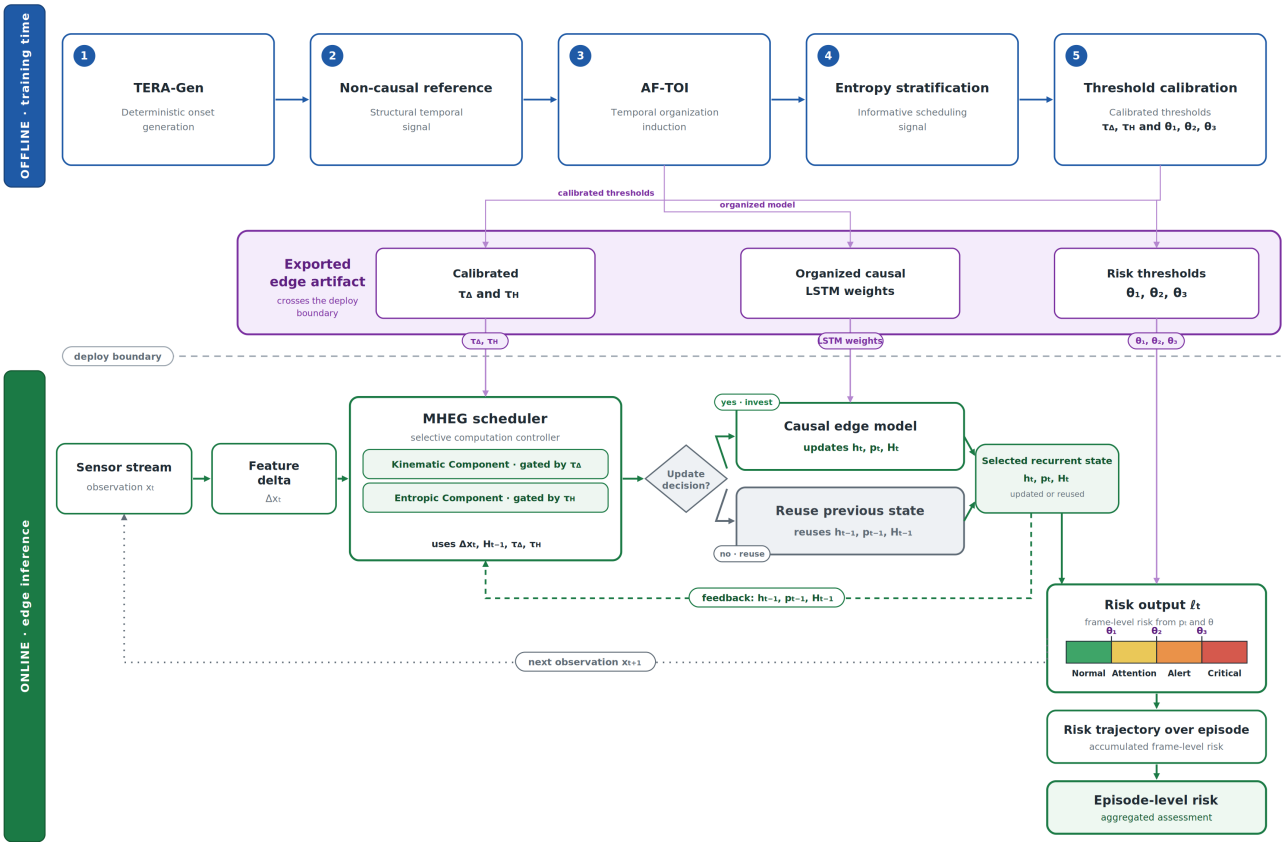


Figure 2: Operational architecture of TERA. Offline stages induce temporally organized trajectories and export the causal LSTM with calibrated thresholds. Online, at each timestep t, MHEG decides, from Δx_t, H_{t-1}, and the thresholds τ_Δ, τ_H, whether to run a full causal-window update or reuse the emitted probability, under coverage-preservation constraints.

is episode duration. TERA can be read as a trustworthy selective computation mechanism for safety-critical vehicular edge environments, with operational formulation

$$\begin{aligned} \min \quad & \mathrm{TTD}_{\mathrm{ef}} \\ \text{subject to} \quad & \mathrm{FR} \leq 0.05 \end{aligned} \tag{2}$$

Equation 2 abstracts the objectives that guide design and evaluation; it is not an optimization problem solved explicitly. Among configurations satisfying $\mathrm{FR} \leq 0.05$, selective computation is not an independent objective but a consequence of the scheduler identifying redundant regions without compromising critical-event detection. The six objectives are auxiliary detection (Balanced F1 [46]), auxiliary calibration (ECE [12]), auxiliary partial timeliness (gross delay TTD_b [47]), the primary effective timeliness $\mathrm{TTD}_{\mathrm{ef}}$ (Eq. 1), the safety constraint $\mathrm{FR} \leq 0.05$ [48], and auxiliary computational economy (selective suppression rate, %Sup [6]).

Sections 3.1–3.4 follow the causal chain from deterministic onset anchoring through temporal organization induction to trustworthy selective computation.

3.1. Deterministic Temporal Anchoring (TERA-Gen)

Controlled induction and frame-precise evaluation presuppose a deterministic onset as reference. In naturalistic datasets, onset is fixed by retrospective annotation with variability of ≈0.31 s at 9.6 fps, which masks real operational differences between schedulers by making it impossible to tell whether the allocator invested resources at emergence windows. TERA-Gen is a controlled laboratory for this, not a substitute for naturalistic data. By anchoring risk targets to SHRP2 odds ratios it preserves epidemiological coherence while removing the variability that would obscure the effect of entropic differentiation on selectivity.

Scenarios span progressive and abrupt risk transitions, enabling frame-precise quantification of $\mathrm{TTD}_{\mathrm{ef}}$, miss rate, and robustness. The system models 31 attributes across visual, kinematic, contextual, and acoustic domains grounded in traffic epidemiology.

Input vector structure The vector $x_t \in \mathbb{R}^{31}$ aggregates four domains: behavioral visual (seven driving classes, lateral and gaze deviation), kinematic (speed, longitudinal accelerations, gyroscopic signals), contextual (road type, precipitation, night driving, traffic density), and acoustic (siren, horn, phone ring, background

Table 2
Unified performance objective matrix of the TERA ecosystem. **Role**: **P** = primary; **S** = safety constraint; **A** = auxiliary evidence. **Meaning** summarizes the physical reading of each metric. † Proposed formulation; see Eq. 1.

Objective / Metric	Symbol	Role	Target	Meaning
Multimodal detection (Balanced F1)	F1	A	Max	Harmonic mean of precision and recall over the stream [46].
Predictive calibration (ECE)	ECE	A	Min	Gap between declared confidence and empirical frequency; improves as a byproduct of organization, not as an objective [12].
Partial timeliness (gross delay)	TTD_b	A	Min	Mean latency over detected episodes only; optimistic, as misses are excluded [47].
Effective timeliness	$\mathrm{TTD}_{\mathrm{ef}}$	**P**	**Min**	$\mathrm{TTD}_b + \mathrm{FR} \times T_{\mathrm{ep}}$ (T_{ep}=10.0 s); adds the miss penalty and arbitrates allocation. Proposed†.
Coverage stability (Failure Rate)	FR	S	≤ 0.05	Proportion of missed critical episodes; an episode is missed when it has no valid detection under the stable-detection rule of Eq. 14 ($p_t \geq \theta_{\mathrm{TTD}}$ for m=3 consecutive frames from the onset) – the same rule used by the threshold calibration. The aggregated episode score over the final k=6 frames defines the episode *classification* consumed by F1, not FR; hard constraint on MHEG calibration [48].
Computational economy (suppression)	%Sup	A	Max	Fraction of frames whose LSTM update is suppressed within the FR constraint [6].

Table 3
Risk behaviors calibrated by SHRP2 [15] and target parameterization.

Behavior	Risk Level	OR	*risk_target*
Normal driving	Normal	1.0	0.050
Hands-free conversation	Attention	1.3	0.074
Mild fatigue / drowsiness	Alert	4.0	0.290
Reaching for moving object	Critical	9.1	0.698
Manual cell phone dialing	Critical	12.2	0.946

noise). Two further indicators encode instantaneous detector reliability.

Epidemiological calibration Risk targets are projected onto $[0.05, 1.00]$ by

$$risk_target \approx 0.05 + \frac{\mathrm{OR} - 1}{12.5} \tag{3}$$

which preserves ordinal severity without claiming to model absolute collision probability. Table 3 gives the resulting values, and the continuous reference label combines mean trajectory and peak intensity,

$$y_{\mathrm{ep}} = 0.7\,\mathrm{mean}(y_{\mathrm{frame}}) + 0.3\,p_{95}(y_{\mathrm{frame}}) \tag{4}$$

Emergency regimes as design principle TERA-Gen builds critical episodes in two ramp patterns. The *progressive* regime (60% of critical scenarios) simulates gradual development with a latent baseline of 8–20 timesteps and a ramp of ≈1.4 s. The *abrupt* regime (the remaining 40%) models fast-ramp perturbations <0.6 s, the worst case for coverage under adaptive computation, so the scheduler is tested exactly where it must allocate resources to a rapidly emerging risk.

Methodological choice of binary supervision Episodes use binary supervision (Normal vs. Critical). Section 4.7 verifies that this onset-anchored supervision suffices to induce continuous ordinal probabilistic trajectories. Continuous SHRP2-calibrated targets generate temporally structured trajectories; training labels classify episodes as Normal or Critical, isolating temporal structuring from refined risk regression.

Deterministic onset is the precondition for induction, because aligning causal trajectories to a structural signal is only meaningful when the onset instant is fixed and variability-free. AF-TOI operates directly on this anchor.

3.2. Temporal Organization Induction (AF-TOI)

AF-TOI induces temporal organization rather than transferring accuracy. It uses onset-oriented reference-to-target structural alignment during offline training, aiming at predictive dynamics that separate stable regions, pre-onset regions, and transition windows, which is distinct from model compression. Once that differentiation emerges, predictive uncertainty stops behaving as a generic confidence estimate and becomes structured by the temporal phase of the episode.

This organization is the enabling property studied here. As trajectories acquire temporal structure, entropy stratifies across episode phases, separating routine intervals from critical transitions, and that stratification is the scheduling signal MHEG depends on. Resource allocation remains a downstream consequence of the induced structure, not a direct output of AF-TOI.

The causal constraint of embedded inference motivates the induction: a unidirectional LSTM sees only accumulated history, so abrupt transitions with sub-0.6 s pre-onset windows risk weakly differentiated uncertainty and insufficient stratification.

To overcome this, AF-TOI uses a non-causal BiLSTM reference with bidirectional access to each episode's full temporal evolution. Free of causal restrictions, the reference develops trajectories that encode the organization of risk transitions across the whole window. AF-TOI transfers this structure to the causal target through offline alignment, so the target reproduces temporally organized dynamics while staying causal at deployment. The induction stays in the development environment, and only the resulting causal model is exported.

Onset-Oriented Temporal Loss Function The causal target is optimized under the composite loss implemented in the released training code, which we state in full because the mechanism-removal ablation of Section 4.6 zeroes specific terms of it:

$$\mathcal{L}_{\text{AF-TOI}} = \alpha(e)\,\mathcal{L}_{\text{hard}} + \beta(e)\,\mathcal{L}_{\text{align}} + \gamma_{\text{ep}}\,\mathcal{L}_{\text{ep}} + \delta(e)\,\mathcal{L}_{\text{onset}} + \lambda_{\text{late}}\,\mathcal{L}_{\text{late}} + \lambda_{\text{pre}}\,\mathcal{L}_{\text{pre}} + \lambda_{\text{tail}}\,\mathcal{L}_{\text{tail}} + \lambda_{\text{neg}}\,\mathcal{L}_{\text{neg}} \tag{5}$$

with γ_{ep}=0.25, λ_{late}=2.0, λ_{pre}=1.5, λ_{tail}=1.2, and λ_{neg}=6.0 (Table 6).

The supervised term is a ramp-weighted, class-weighted binary cross-entropy over the frame head (positive frames reweighted by the inverse class frequency, as implemented),

$$\mathcal{L}_{\text{hard}} = \frac{1}{T}\sum_{t=1}^{T} \omega_e(t)\,\text{BCE}(\hat{p}_t,\, y_{\text{frame},t})\,, \qquad \omega_e(t) = 1 + \lambda_{\text{ramp}}(e)\,\tfrac{t}{T-1}, \tag{6}$$

with λ_{ramp} rising from 0.20 to 1.00 over training, and $\mathcal{L}_{\text{ep}}$ the analogously class-weighted BCE on the episode head. The reference (teacher) enters through a single term: a temperature-scaled alignment in probability space, restricted to frames of critical episodes,

$$\mathcal{L}_{\text{align}} = \underset{t\,\in\,\text{critical}}{\text{mean}} \Big(\sigma\big(z_s(t)\big) - \sigma\big(\tfrac{z_r(t)}{T}\big)\Big)^2, \qquad T = 2.0, \tag{7}$$

with $z_r(t)$ the reference logits and $z_s(t)$ the causal-target logits (a probability-space analogue of temperature-scaled distillation, chosen over the KL form for numerical stability under the trajectory-shaping terms below).

All remaining terms are *onset-anchored trajectory-shaping penalties on the student's own output*: they depend on the deterministic onset t_0 of TERA-Gen and not on the reference. Over the post-onset window $\mathcal{W} = [t_0,\, t_0{+}K]$, K=50, with the normalized exponential focus

$$w_\alpha(t) = \frac{|\mathcal{W}|\; e^{-\alpha(t-t_0)}}{\sum_{u\in\mathcal{W}} e^{-\alpha(u-t_0)}}, \qquad t \in \mathcal{W}, \tag{8}$$

the composite $\mathcal{L}_{\text{onset}} = 0.60\,\mathcal{L}_{\text{track}} + 1.60\,\mathcal{L}_{\text{floor}} + 1.20\,\mathcal{L}_{\text{mono}} + 1.50\,\mathcal{L}_{\text{push}}$ collects the onset-window tracking penalties: w_α-weighted squared tracking of a target curve $g(t) = 0.30 + 0.60\,\tilde{y}(t)$ built from the soft frame labels, a hinge floor at $\max(p_{\min}(e),\, 0.95\,g(t))$ with $p_{\min}$ annealed from 0.50 to 0.78, a monotonicity hinge requiring frame-to-frame rises of at least 0.015 plus the target increment, and an initial push of the first eight frames toward a ramp from 0.45 to $\min(0.80,\, p_{\min}{+}0.10)$. $\mathcal{L}_{\text{late}}$ is the post-onset asymmetric penalty, a w_α-weighted hinge $\text{relu}(\theta_{\text{late}} - \hat{p}_t)$ on $\mathcal{W}$ with θ_{late}=0.10, active once $\delta(e) > 0.5$ (from the ramp-up phase); $\mathcal{L}_{\text{pre}}$ is the pre-onset push, a hinge toward a rising ramp ($0.20 \to 0.35$) on $[t_0{-}8,\, t_0)$, applied to progressive critical episodes only and active once $\delta(e) > 1.0$ (late refinement); $\mathcal{L}_{\text{tail}}$ floors the last six frames of critical episodes at 0.40; and $\mathcal{L}_{\text{neg}}$ caps the last six frames of non-critical episodes at 0.04.

The mechanism-removal ablation of Section 4.6 zeroes exactly $\mathcal{L}_{\text{onset}}$, $\mathcal{L}_{\text{late}}$, and $\mathcal{L}_{\text{pre}}$ – the onset-anchored terms – leaving every other term intact. A note on symbol reuse: T in Eq. 7 is the alignment temperature (T=2.0), distinct from the episode length T; K=50 in the window $\mathcal{W}$ is the post-onset horizon, distinct from the kinematic channel set $\mathcal{K}$ of Eq. 10; and α in Eq. 8 is the exponential-focus decay (α=0.15), distinct from the supervised weight $\alpha(e)$. This decomposition also states the relation to temporal knowledge distillation precisely: the teacher contributes only the probability-space alignment of Eq. 7, while the temporal organization is imposed by onset-anchored shaping of the student's own trajectory, Eqs. 8 and the associated penalties – the closest published relatives being exponentially weighted anticipation losses for maneuver prediction rather than distillation objectives.

Three-Phase Structural Alignment Scheduler Direct coupling to the reference signal in early epochs can cause gradient instability, so the loss weights follow a three-phase schedule (warm-up E_{warm}=12 epochs, ramp-up until E_{ramp}=28, refinement until E_{total}=80): the alignment weight rises as

$$\beta(e) = \begin{cases} 0, & 1 \le e \le E_{\text{warm}} \\ \beta_{\max}\,\dfrac{e - E_{\text{warm}}}{E_{\text{ramp}} - E_{\text{warm}}}, & E_{\text{warm}} < e \le E_{\text{ramp}} \\ \beta_{\max} = 0.35, & E_{\text{ramp}} < e \le E_{\text{total}} \end{cases} \tag{9}$$

while the supervised weight $\alpha(e)$ decays from 1.0 to a floor of 0.50 (supervision is never switched off) and the onset-shaping weight $\delta(e)$ grows from 0.30 to 2.80, shifting emphasis from label fidelity to temporal organization as training progresses.

The scheduler is a **constitutive** part of AF-TOI; removing it compromises coupling stability and the quality of the induced differentiation, as the architectural ablation indicates (Section 4.9).

The resulting stratification, with stable intervals at low entropy and pre-onset and transition windows at

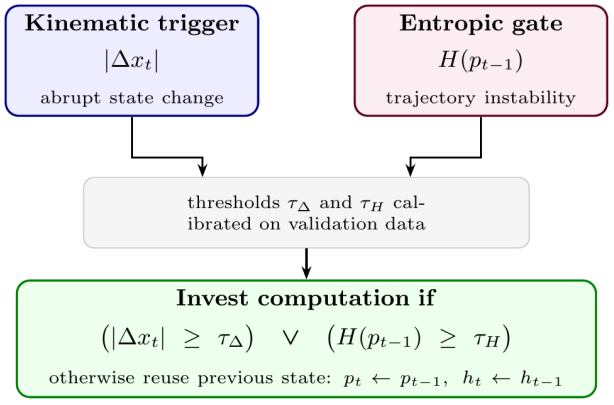


Figure 3: MHEG runtime computation scheduler structure. The kinematic component is an unconditional resource allocation trigger that forces LSTM computation whenever physical state changes abruptly, regardless of entropic state. The entropic component is the trajectory-dependent scheduling signal. When temporal organization is absent, entropy remains undifferentiated and the scheduler loses temporal discrimination power; when induced by AF-TOI, entropy stratifies near onset and the scheduler can allocate resources to emergence windows while safely suppressing redundant intervals.

elevated, varying uncertainty, is the scheduling signal exploited next.

3.3. Runtime Selective Computation (MHEG)

MHEG reads the stratification that AF-TOI produces; it does not create it. That dependence is deliberate and consequential: the scheduler's usefulness is inherited from the offline-induced organization, not from the gating rule on its own. When the stratification is absent, the entropic component has no basis for discrimination, which is precisely the failure reproduced by the negative control.

MHEG is the runtime scheduler and adaptive inference controller of the embedded system. At each timestep it decides whether to run a full causal-window update or reuse the emitted probability (Section 3.4), using the AF-TOI-induced organization. Its logic rests on two components, synthesized in Figure 3.

Kinematic Component as Unconditional Resource Allocation Trigger

$$|\Delta x_t| = \max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \tag{10}$$

Here $\mathcal{K}$ denotes the set of monitored kinematic channels and $K = |\mathcal{K}|$ its cardinality ($K \leq D$). When this variation exceeds τ_Δ, the LSTM core runs regardless of entropic state. The unconditional trigger guarantees allocation in abrupt episodes with sub-0.6 s pre-onset windows, where entropic differentiation may not manifest before the transition begins.

Entropic Component as Trajectory-Dependent Scheduling Signal

$$H(p_{t-1}) = -p_{t-1}\log_2 p_{t-1} - (1-p_{t-1})\log_2(1-p_{t-1}) \tag{11}$$

(with $0 \log 0 \equiv 0$; the implementation clips p to $[10^{-7}, 1-10^{-7}]$). When entropy exceeds τ_H, the scheduler invests full LSTM processing. Here entropy is a decision variable reflecting trajectory instability, not an error indicator. The scheduler does not need globally calibrated confidence; it needs uncertainty with temporal differentiation. Absent global calibration does not compromise allocation, but absent temporal differentiation does.

The discriminative power of the entropic component depends on trajectory organization. Without AF-TOI-induced differentiation, entropy sits in a narrow range and the scheduler loses selectivity; with it, phases stratify along the entropic axis and the criterion gains temporal discrimination. This dependency is the causal relationship at the center of the hypothesis, verified in Section 4.

Joint Threshold Calibration With causal-target weights frozen, a per-seed grid search over the validation set selects the threshold combination that maximizes suppression while keeping FR ≤ 0.05. Calibration uses TERA-Gen synthetic data; online re-calibration is a necessary extension for production.

Scheduling Criterion via Logical Disjunction

$$\max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \geq \tau_\Delta \quad \vee \quad H(p_{t-1}) \geq \tau_H \tag{12}$$

When neither condition holds, the LSTM update is suppressed and previous values propagate. The disjunction reflects the asymmetric cost of safety-critical edge systems, where suppressing computation at a transition window is far costlier than a redundant inference step.

Operational Regimes Threshold magnitude determines three regimes. As thresholds $\to 0$, the system runs full continuous inference: nothing is suppressed, cost is maximal, FR is zero. As thresholds $\to \infty$, suppression is total and equivalent to halting all LSTM updates: cost is zero, FR is maximal. Between these, per-seed calibration on validation data yields the trustworthy hybrid regime the ecosystem occupies, maximizing economy under FR ≤ 0.05. The next subsection describes online execution.

3.4. Runtime Inference and Risk Assessment

Algorithm 1 formalizes runtime execution at the edge device. The continuous probability p_t is mapped

Algorithm 1 Efficient inference cycle of the MHEG runtime computation scheduler

Require: Weights θ_{LSTM}; window length W=96 (prefix-window with left-padding; Section 4.1); thresholds τ_Δ, τ_H, θ_1=0.15, θ_2=0.38, θ_3=0.50; h=**0** at the start of each recomputed window; p_0=0; x_0=**0** (the t=1 disjunct forces a full update on the first frame, as implemented)

Ensure: Level $\ell_t \in \{0,1,2,3\}$ emitted at each timestep t

1: **for** each frame x_t **do**
2: $\quad\Delta x_t \leftarrow \max_{k\in\mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}|$
3: $\quad H_t \leftarrow H(p_{t-1})$
4: $\quad$**if** $(t=1)$ **or** $(\Delta x_t \geq \tau_\Delta)$ **or** $(H_t \geq \tau_H)$ **then**
5: $\quad\quad p_t \leftarrow \text{LSTM}(\text{window}_W(x_{1..t});\, \theta_{\text{LSTM}})$ *[recompute causal window (evaluated semantics); single-step variant: Section 3.4]*
6: $\quad$**else**
7: $\quad\quad p_t \leftarrow p_{t-1}$ *[Reuse emitted probability – suppress computation]*
8: $\quad$**end if**
9: $\quad\ell_t \leftarrow \mathbf{1}[p_t \geq \theta_3]\cdot 3 + \mathbf{1}[\theta_2 \leq p_t < \theta_3]\cdot 2 + \mathbf{1}[\theta_1 \leq p_t < \theta_2]\cdot 1$
10: $\quad$**Emit** ℓ_t
11: **end for**

to four discrete risk levels.

$$\ell_t = \begin{cases} 0 \text{ Normal}, & p_t < 0.15 \\ 1 \text{ Attention}, & 0.15 \leq p_t < 0.38 \\ 2 \text{ Alert}, & 0.38 \leq p_t < 0.50 \\ 3 \text{ Critical}, & p_t \geq 0.50 \end{cases} \tag{13}$$

The per-timestep complexity of Algorithm 1 is dominated by line 5. Lines 2–4 and 8–10 cost $\mathcal{O}(K)$ and $\mathcal{O}(1)$, respectively (a maximum over the K monitored kinematic channels of Eq. 10, one closed-form binary entropy, and three threshold comparisons), and involve no matrix operation. One recurrent step of the two stacked layers costs $\mathcal{O}(4(DH{+}H^2){+}8H^2)$ (the second layer's input has dimension H), i.e., ≈57.2 K MACs for D=31, H=64. The window recomputation of line 5 runs W=96 such steps, so an executed update costs ≈5.49 M MACs and fires only when the disjunction of line 4 holds; the expected per-frame cost of Algorithm 1 is therefore $c_{\text{gate}}{+}(1{-}s)\,W\,\mathcal{O}(4(DH{+}H^2){+}8H^2)$, where s is the suppression rate (%Sup) and $c_{\text{gate}} = \mathcal{O}(K)$ the gate overhead, constant for a deployment with fixed K. The worst case (no suppression) is one full window per frame plus $\mathcal{O}(K)$; the best case is the gate alone. The single-step formula without the W factor, $c_{\text{gate}}{+}(1{-}s)\,\mathcal{O}(4(DH{+}H^2){+}8H^2)$, applies only to the non-validated persistent-state variant discussed below.

Algorithm 1 states the semantics that every scheduled result in this paper was evaluated under: the causal prefix window of the last W=96 available frames is left-padded with the first frame, the recurrent state is reinitialized at the start of each recomputed window, the inputs of previously suppressed frames are re-read by later updates (the input window must therefore be buffered), and a suppressed frame reuses only the emitted probability. The ms/frame figures of Tables 8 and 22 are wall-clock times of exactly this execution. A cheaper *persistent-state single-step* variant – one recurrent step (≈57.2 K MACs) per executed update, suppressed frames never processed, per-frame cost $c_{\text{gate}} + (1-s)\,\mathcal{O}(4(DH{+}H^2){+}8H^2)$ – is the natural deployment target but is *not validated here*: with the same frozen seed-42 checkpoints and thresholds it agrees with the evaluated semantics on FR under the fixed policy (TTD_b differs by 6 ms, a padding warm-up effect) but is not equivalent under suppression (TTD_{ef} = 0.170 s at 44.4% suppression, against 0.023 s at 44.7% for the evaluated semantics), so it would require its own threshold calibration and evaluation. A verification script reproducing this comparison accompanies the artifact.

3.5. Computational Complexity and Edge Deployment

This subsection checks whether the ecosystem is compatible with resource-constrained deployment under strict offline–online separation.

Table 4 summarizes the computational profile and the separation between training and device. The highest-cost modules (TERA-Gen, the non-causal BiLSTM reference, AF-TOI, and MHEG calibration) stay offline. At runtime only the causal LSTM target, the MHEG scheduler, the recurrent state, and the calibrated thresholds are active on device.

The hidden and cell buffer needs exactly 1,024 bytes (the h and c states of the two layers, $2\times2\times64$ FP32 values – the recurrent state only, not the total runtime memory). The causal LSTM with two layers and H=64 hidden units has 58,242 parameters – $4(DH{+}H^2{+}2H)$ for the first layer (D=31), $4(2H^2{+}2H)$ for the second, whose input has dimension H, and $2(H{+}1)$ for the two linear heads (≈228 KB in FP32, ≈57 KB in INT8), with expected compatibility with STM32H7, Cortex-M55 with Helium, Jetson Nano, and Qualcomm SA8155P. Because every weight is used once per timestep, one recurrent step costs ≈57.2 K multiply–accumulate operations (MACs), i.e., ≈115 KFLOPs under the 1 MAC = 2 FLOPs convention adopted throughout; a verification script that recomputes these counts from the released checkpoints accompanies the artifact.

Table 4
Computational complexity profile of TERA ecosystem components.

Component	Functional role	Complexity	Executes at	On device
TERA-Gen	Deterministic onset episode generation	$\mathcal{O}(E \cdot T \cdot D)$	Training only	✗
Non-causal BiLSTM reference arch.	Source of temporal structural signal for AF-TOI alignment	$\mathcal{O}(8(DH+H^2))$/step	Training only	✗
AF-TOI	Temporal organization induction in causal target architecture	$\mathcal{O}(T)$ per episode	Training only	✗
MHEG calibration	Per-seed grid search of scheduler thresholds	$\mathcal{O}(L \cdot E_{\text{val}} \cdot T)$	Training only	✗
Causal LSTM (embedded target arch.)	On-device recurrent inference	$\mathcal{O}(4(DH+H^2)+8H^2)$/step; $W\times$ per update‡	Continuous runtime	✓
MHEG (runtime)	Runtime computation scheduler / adaptive inference controller	$\mathcal{O}(K)$/step	Continuous runtime	✓
Recurrent state (h, c)	Reinitialized at each window recomputation (evaluated semantics); persistent only in the non-validated single-step variant	$\mathcal{O}(H)$	Continuous runtime	✓
Thresholds $(\tau_\Delta, \tau_H, \boldsymbol{\theta})$	Constant scalar scheduling parameters	$\mathcal{O}(1)$	Continuous runtime	✓

Legend: E: training episodes; T: episode temporal length; D: input dimensionality; H: LSTM hidden dimension; K: number of monitored kinematic channels ($K \leq D$); L: number of combinations evaluated in grid search; ‡evaluated windowed semantics (Section 3.4): each executed update recomputes the causal W-frame window.

4. Experimental Evaluation

Validation follows six movements that track the hypothesis, each answering one question and preparing the next. Movements 1 and 2 examine the mechanism: whether AF-TOI observably induces temporal organization in the causal target, and whether that organization confers scheduling selectivity to MHEG. Movement 3 quantifies the operational consequence on coverage, $\mathrm{TTD}_{\mathrm{ef}}$, and efficiency. The last three probe the nature of the effect: whether it comes from a reorganized failure distribution, whether it is stable across seeds and regimes, and whether it extends to intermediate risk. Architectural ablation and embedded viability close the section.

4.1. Experimental Protocol

The protocol lets temporal organization be induced, validated, and evaluated with frame-level precision, which is what makes the chain from organization to trustworthy selective computation testable rather than assumed: it renders the entropic differentiation the scheduler depends on directly observable against a deterministic onset.

Synthetic multimodal dataset TERA-Gen synthesizes episodes with 13 training recipes, N01–N03 for normal driving and C01–C10 for critical risk. Six exclusive recipes form the borderline set, A01–A03 for the Attention level and L01–L03 for Alert. Training recipes were generated for the main experimental pipeline; borderline recipes were generated independently and never exposed during training, serving exclusively as a held-out evaluation set for Movement 6 (Section 4.7). Table 5 summarizes the full recipe taxonomy.

The dataset consolidates 12,480 episodes across twelve seeds (42–53); the four initial seeds (42–45) of the factorial study were verified to reproduce their four-seed campaign exactly before the extension to twelve. Each seed follows the 70/15/15% split. Each sequence has $T = 96$ timesteps at $\approx$9.6 fps, $D = 31$, and deterministic onset t_0 at frame-level precision. That onset makes the induced organization verifiable, since it fixes the reference against which entropic differentiation and scheduler selectivity are measured. Table 6 details the full parameterization for reproducibility.

Threshold configuration and calibration Thresholds τ_Δ and τ_H are calibrated per seed via grid search on the validation partition, selecting the configuration that maximizes suppression under $\mathrm{FR} \leq 0.05$. A detection is valid only when $p_t \geq \theta_{\mathrm{TTD}}$ for $m = 3$ consecutive timesteps; the reported detection instant t_{det} is the *first* frame of that run (a retrospective marker), while causal confirmation occurs at its third frame, so a causal alarm would fire $(m-1)\Delta t \approx 0.21$ s after the reported instant.

Table 5
Recipes used in TERA-Gen synthetic episode generation.

Set	Level	Recipes	Application
Training	Normal	N01–N03	Train, validation, test
	Critical	C01–C10	Train, validation, test
Borderline	Attention	A01–A03	Evaluation only
	Alert	L01–L03	Evaluation only

$$\begin{aligned} \mathrm{TTD}_{\mathrm{det}} &= (t_{\mathrm{det}} - t_0) \cdot \Delta t, \\ t_{\mathrm{det}} &= \min\{t \geq t_0 \mid p_t, p_{t+1}, p_{t+2} \geq \theta_{\mathrm{TTD}}\} \end{aligned} \tag{14}$$

$$\mathrm{TTD}_{\mathrm{ef}} = \mathrm{TTD}_{\mathrm{det}} + \mathrm{FR} \times T_{\mathrm{ep}}, \qquad T_{\mathrm{ep}} = 10.0\,\mathrm{s}, \tag{15}$$

where $\mathrm{TTD}_{\mathrm{det}}$ is the mean of Eq. 14 over the *detected* critical episodes and FR is the proportion of missed critical episodes; this is the single definition used by every table in this paper, identical to Eq. 1. In the per-episode distributions of Figures 8 and 9, a missed episode is displayed at the penalty value T_{ep}; the aggregate statistic reported in the tables is the expression above, in which the miss penalty enters through FR rather than through per-episode imputation (the mean of the displayed per-episode values equals $(1 - \mathrm{FR})\,\mathrm{TTD}_{\mathrm{det}} + \mathrm{FR}\,T_{\mathrm{ep}}$ and is therefore not the tabulated quantity).

Infrastructure and significance tests Latency is measured in unit-batch mode ($N_{\mathrm{batch}} = 1$) on an NVIDIA RTX 3050 Ti GPU with PyTorch 2.0 and 100-iteration warmup, per frame of the prefix-window streaming simulation (each executed update runs one full causal window; Section 3.4). Hypotheses use the paired Wilcoxon test [49], with effect size quantified by the matched-pairs rank-biserial correlation [50], computed on the signed ranks of the nonzero paired differences, and 95% confidence intervals by recipe-stratified bootstrap [51]. The endpoints form an explicit hierarchy: $\mathrm{TTD}_{\mathrm{ef}}$ is the primary operational endpoint; FR acts as a safety constraint ($\mathrm{FR} \leq 0.05$ per seed, Section 2.2); F1, ECE, %Sup, and the gross latencies are auxiliary descriptors. Inferential claims are confined to the twelve-seed campaigns (the factorial study and the retrained mechanism-removal ablation of Table 11); the complementary campaigns (n=4) are reported descriptively, and the per-duration contrasts of Section 4.12 are exploratory, with Holm correction applied within each endpoint family.

Experimental protocol map Because the campaigns that follow deliberately use different protocols, Table 23 (placed at the end of the evaluation, after Table 22) maps every experiment to its seeds, model provenance, decision rule, and ECE variant; magnitudes are comparable within a row, not across rows.

Table 6
Architecture, optimization, and temporal organization induction hyperparameters.

Component	Symbol / Variable	Value	Role in Pipeline
General Architecture and Data	D (input_dim)	31	Multimodal vector x_t dimensionality
	H (hidden_dim)	64	Hidden units per recurrent layer
	num_layers	2	Stacked recurrent layers
	dropout	0.1	Structural regularization factor
Non-Causal BiLSTM Reference Arch. (*Offline*)	bi	True	Enables non-causal bidirectional processing
	factor	2	Hidden state multiplier ($2 \times H = 128$)
	lr	0.001	Adam optimizer learning rate
	epochs	60	Training epochs with combined $\mathcal{L}_{\text{ref}}$ loss
Loss Function $\mathcal{L}_{\text{AF-TOI}}$ (Causal Target Arch.)	α (onset_exp_alpha)	0.15	Post-t_0 decay rate (Eq. 8)
	T (temp)	2.0	Alignment smoothing temperature (Eq. 7)
	$\beta_{\max}$ (beta_max)	0.35	Peak alignment weight (Eq. 9)
	γ_{ep}; $[\delta_{\text{start}}, \delta_{\text{end}}]$	0.25; [0.30, 2.80]	Episode-head weight; onset-shaping weight range (Eq. 5)
	K; N_{pre}	50; 8	Post-onset window and pre-onset push window (frames)
	λ_{pre}; λ_{tail}; λ_{neg}	1.5; 1.2; 6.0	Pre-onset push, critical-tail floor, non-critical cap weights
	λ_{late} (lambda_late)	2.0	Late Penalty multiplier ($\mathcal{L}_{\text{late}}$)
	θ_{late} (theta_late)	0.10	Causal target reactivity threshold
Structural Alignment Scheduler	E_{warm} (warmup_epochs)	12	Initial epochs with $\beta(e)$=0 (no alignment term; the fixed-weight and onset-shaping terms remain active at their initial weights)
	$E_{\text{ramp}} - E_{\text{warm}}$ (rampup_epochs)	16	Linear growth window of $\beta(e)$ (Eq. 9)
	E_{total} (epochs)	80	Total causal target optimization epochs
	$[\lambda_{\text{start}}, \lambda_{\text{end}}]$	[0.2, 1.0]	Hard-loss temporal ramp bounds (Eq. 6)

4.2. Movement 1. Temporal Mechanism

Movement 1 checks, at frame level, whether AF-TOI observably reorganizes the causal target's internal trajectory structure and thereby gives MHEG a temporally discriminative signal. This mechanistic evidence precedes any attribution of allocation gains to organization.

In the Baseline, H_t stays poorly differentiated across driving phases: phase distributions largely overlap, with no clear separation between stable and transition zones (Figure 4(a)). The entropic scheduling signal is then undiscriminating at the periods of highest criticality, providing no temporal separation between emergence windows and redundant intervals; only the unconditional kinematic trigger remains protective. With AF-TOI, phases stratify along the entropic axis (Figure 4(b)): Normal anchors at low entropy (density peak ≈0.35 bit), Pre-onset shifts upward toward the transition regime (median ≈0.94 bit), and Post-onset accumulates at the maximum (1 bit). This is the pattern that gives MHEG a usable scheduling signal.

As a byproduct of the induction, alignment between declared confidence and empirical frame-level accuracy improves, with $\text{ECE}_{\text{replay}} = 0.044$ for AF-TOI versus 0.129 for Baseline (Figure 4(c); replay logs of seed 42, not comparable to the pipeline-level ECE of Table 8). This calibration gain accompanies stratification but is not the objective.

Figure 5 adds the episode-level view. In AF-TOI+MHEG, suppressed intervals concentrate in low-entropy periods while LSTM computation is preserved near onset, so the scheduler invests at emergence windows and reuses buffers at redundant timesteps. In the Baseline, the residual onset protection comes from the kinematic trigger alone – 37% of onset-window frames are still suppressed, against 10% under AF-TOI– and the entropic component contributes no temporal discrimination.

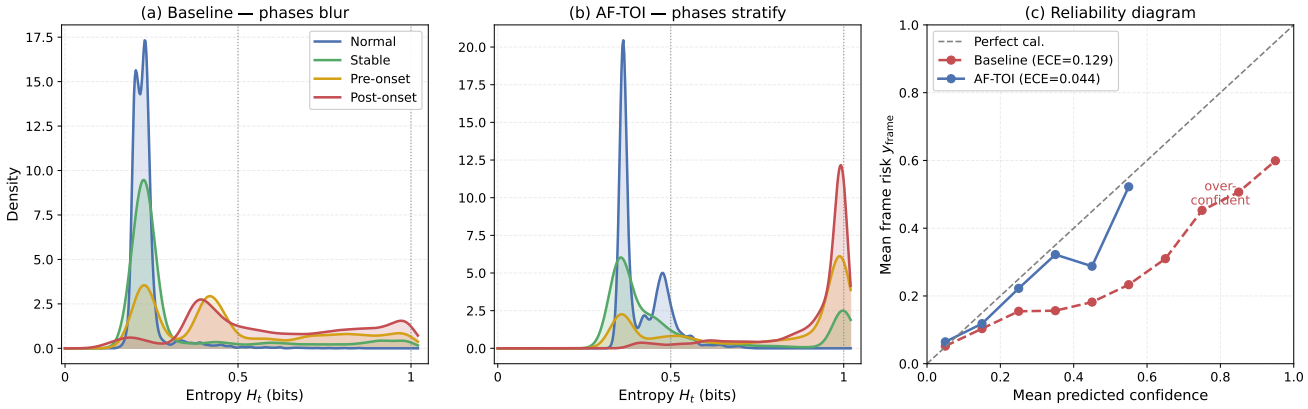


Figure 4: Trajectory organization across driving phases for seed 42. Panels compare Baseline entropy structure, AF-TOI-induced phase stratification, and frame-level calibration alignment ($\mathrm{ECE}_{\mathrm{replay}} = 0.044$ vs. 0.129 for Baseline; entropy axes in bits, the unit of Eq. 11).

Concretely, each panel of Figure 5 tracks one episode of seed 42 frame by frame: the solid curve is the emitted probability p_t, the dashed curve the entropy H_t (in bits) that the gate reads, the vertical line the deterministic onset t_0, the marker ∇ the detection instant, and the shaded spans the frames on which MHEG reused the buffer instead of updating. The exemplars are selected by a pre-declared neutral criterion: within each regime, the episode whose pre/post-onset entropy contrast is the *median* among the critical episodes of seed 42 with a centrally located onset (progressive: episode 36, t_0=5.42 s; abrupt: episode 148, t_0=2.50 s). Reading the AF-TOI+MHEG column: entropy rises from ≈0.5 to ≈1.0 bit across the progressive transition (from 0.36 to 0.73 bit in the abrupt one), and the fraction of suppressed frames drops from 58% before the onset to 14% (progressive) and 43% (abrupt) after it – computation is preserved where the transition happens. In the Baseline column the entropy profile stays comparatively flat (0.49 → 0.56 and 0.24 → 0.52 bit) and suppression is nearly unchanged across the onset (55–56%): the entropic gate has no temporal criterion there, consistent with the aggregate profile over the 120 critical episodes of Figure 6. These exemplars do not establish aggregate latency differences: detection markers of single exemplars do not arbitrate latency – in both exemplars the Baseline trace happens to cross θ at the onset itself – so aggregate latencies are settled by Table 8, not by this figure. And a minority of abrupt critical episodes (12 of 48 in this seed) runs near maximal entropy throughout, with $p \approx 0.5$ already before the onset; that saturated regime is visible in Figure 6 and is revisited in Movement 6.

Under AF-TOI, H_t responds more sensitively to risk transitions, acting as a reliable scheduling variable. Movement 2 tests whether this translates into scheduler selectivity.

4.3. Movement 2. Scheduler Selectivity

If the reorganization is operationally relevant, MHEG should concentrate investment at the highest-uncertainty windows and reuse buffers at low-entropy timesteps.

Figure 5 already shows the resulting asymmetry: with AF-TOI, suppressed intervals concentrate in pre-onset periods while the onset neighborhood is consistently preserved, so computation goes to emergence windows in both regimes. In the Baseline, the entropic criterion has no temporal semantics; what onset protection remains (37% of onset-window frames still suppressed) is supplied by the kinematic trigger.

Figure 6 extends this over 120 critical sequences from seed 42. In the Baseline, H_t stays nearly flat across the onset (0.48 → 0.55 bit on average), giving the gate no discrimination signal. With AF-TOI, H_t rises to 0.95 bit around the onset, shifting MHEG's investment to emergence timesteps: over the critical episodes of this seed, only 9.8% of onset-window frames are suppressed, against 34.3% elsewhere (Baseline: 37.2% vs. 54.4%). The alignment between entropic elevation and onset proximity is how the scheduler acquires temporal discrimination.

The evidence is consistent with scheduler selectivity tracking trajectory organization: present, it concentrates investment at emergence timesteps; absent, it operates without a temporal criterion.

4.4. Movement 3. Operational Consequence

The 2×2 factorial study isolates the operational consequence on the metrics that arbitrate the central conflict. The negative control is the most informative point, measuring what happens to allocation when selective computation is applied to unorganized trajectories. Table 7 details the design; global results are in Table 8.

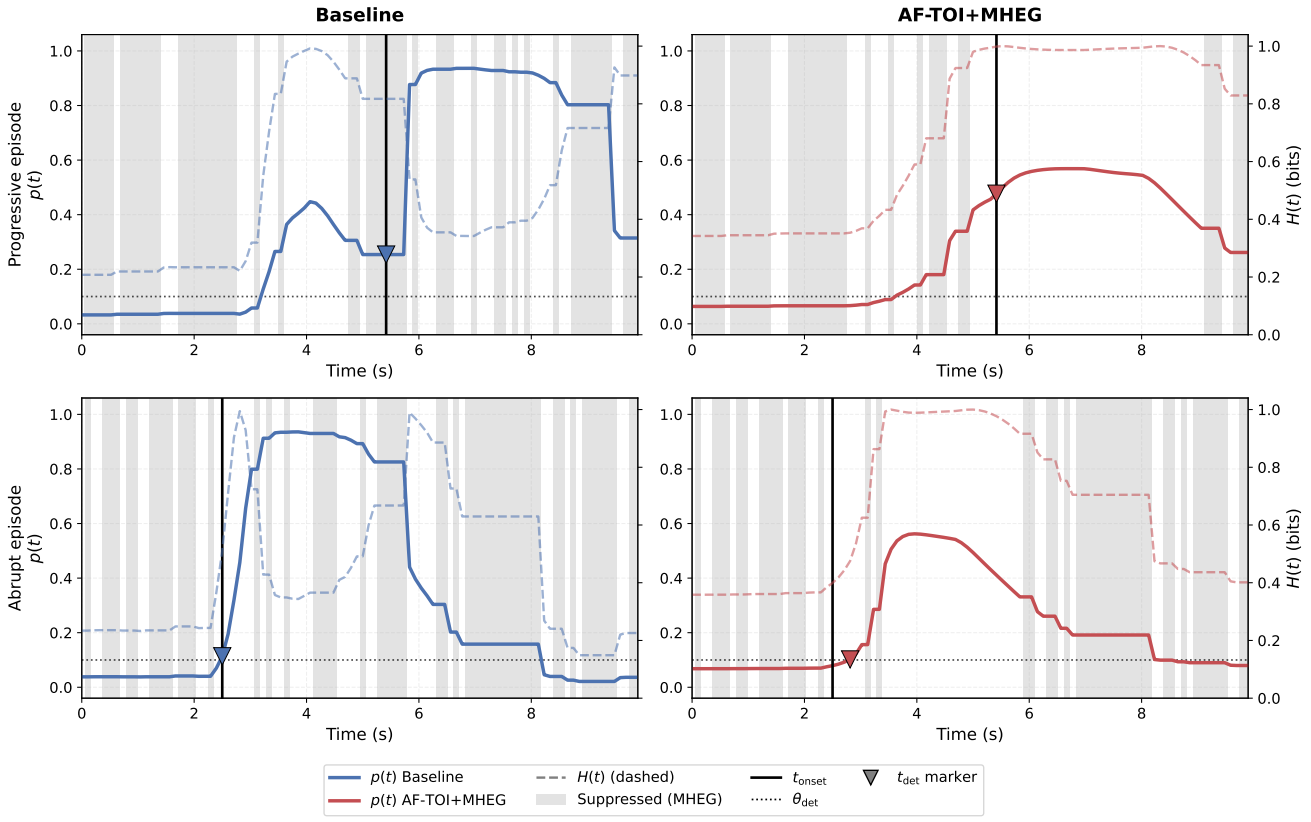


Figure 5: Frame-by-frame temporal evidence for seed 42. Columns show the Baseline and AF-TOI+MHEG configurations; rows show progressive and abrupt episodes. Solid curves denote p_t, dashed curves denote H_t (bits), vertical lines mark t_{onset}, markers $\triangledown$ indicate t_{det}, and gray regions indicate MHEG buffer reuse. Exemplars are the median pre/post-onset entropy-contrast episodes per regime among seed-42 critical episodes with central onset (progressive: episode 36, t_0=5.42 s; abrupt: episode 148, t_0=2.50 s); the selection criterion was fixed before inspection.

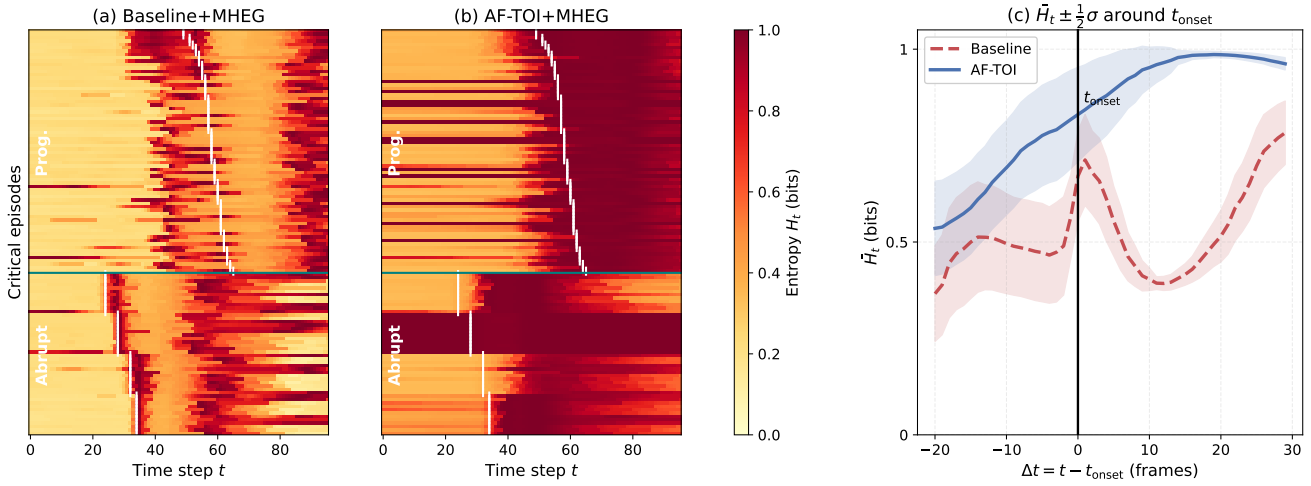


Figure 6: Temporal H_t profile across 120 critical episodes for seed 42. The heatmaps compare Baseline and AF-TOI entropy trajectories around the deterministic onset reference.

Table 7
2×2 factorial design and per-configuration seed coverage.

Configuration	AF-TOI	MHEG	Seeds	Methodological Role
Baseline Fixed	✗	✗	12/12	Basal reference without temporal organization
Baseline+MHEG	✗	✓	12/12	**Negative control: unsafe scheduling without temporal org.**
AF-TOI Fixed	✓	✗	12/12	Temporal organization isolated; no scheduler
AF-TOI+MHEG	✓	✓	12/12	Complete ecosystem: trustworthy selective computation

Seeds: number of seeds (42–53) producing valid runs per configuration.

The negative control as unsafe adaptive computation Applying MHEG to unorganized trajectories has non-neutral effects on allocation quality. Baseline+MHEG holds FR = 0.035 ± 0.053 at the fixed-baseline level, so it does not reduce misses, and it still degrades $\mathrm{TTD}_{\mathrm{ef}}$ by 13.9%, from 0.500 s to 0.570 s. Without organization, the entropic component provides no temporal discrimination, and suppression also lands on emerging-risk timesteps – in the replay logs of seed 42, the Baseline gate suppresses 37% of onset-window frames against 54% elsewhere, whereas the organized gate suppresses 10% against 34%. Under the frozen operating policy this shows up as degraded effective delay; at strictly budget-matched cost the corresponding contrast is the FS@onset diagnostic of Section 4.10, and its temporal-scale limits are stated in Section 4.12. This is the regime that fails the per-seed coverage criterion of Section 2.2: apparent per-frame cost drops with no coverage guarantee while the effective delay worsens. The result points to the absence of the enabling property, not a scheduler design flaw.

The complete ecosystem and trustworthy selective computation Under AF-TOI organization, MHEG operates on differentiated trajectories. The miss rate is preserved at 0.001 ± 0.005, with $\mathrm{TTD}_{\mathrm{ef}}$ rising only 0.016 s relative to AF-TOI fixed. The latency impact of scheduling is about 4.3× smaller in the complete ecosystem than in the negative control (0.016 s vs. 0.070 s), and absolute $\mathrm{TTD}_{\mathrm{ef}}$ with active scheduling is 0.099 s versus 0.570 s. In this synthetic scenario, that convergence is consistent with efficiency and coverage coexisting once the scheduler has temporally organized trajectories.

Regime asymmetry in progressive episodes AF-TOI Fixed anticipates progressive onsets by $A_p = 1.200 \pm 0.437$ s on average, against 0.657 ± 0.325 s for the Baseline: on slowly developing episodes the organized trajectory crosses the detection threshold well before the deterministic onset. Because the global TTD_b of Table 8 measures only the post-onset delay of detected episodes, an episode whose trajectory is still above threshold at the onset enters it with zero delay, while an early activation that fades before the onset re-enters with its own post-onset delay – both patterns occur in the seed-42 data. This is why the global means (0.069–0.223 s) sit below the per-regime figures: A_p and TTD_b are different quantities on different temporal references, not two aggregations of the same measure. The coverage benefit concentrates in the abrupt regime, where the miss rate falls from 0.082 to 0.004, exactly where misses cost most.

TTD_b vs. TTD_{ef} Gross delay counts only detected episodes and reads optimistically because it hides misses. $\mathrm{TTD}_{\mathrm{ef}}$ corrects this for edge systems where coverage across all episodes matters more than mean latency on successful detections.

Statistical analysis The AF-TOI F1-Score (0.846 ± 0.180, CV ≈21%) shows greater inter-seed variance than Baseline (0.366 ± 0.069), reflecting sensitivity of the alignment process to initialization.

To address the limited inferential resolution of the initial four-seed campaign, the factorial study covers twelve seeds (42–53): an initial four-seed campaign (42–45) extended by eight further seeds. The eight additional seeds were fixed before inspecting their outcomes and before recomputing any inferential statistics for the expanded twelve-seed campaign; the sample size was not adjusted as a function of the additional-seed results, and all twelve are reported regardless of outcome; the expansion was executed with the same frozen pipeline (identical configuration and generator hashes), and the four initial seeds were verified to reproduce their four-seed campaign exactly before integration. The twelve-seed campaign covers the factorial study and the temporal-scale generalization analysis. The secondary analyses run on the initial four seeds (42–45), for two distinct reasons: the confidence-threshold and smoothed-entropy gates and the calibration diagnostics *replay* stored per-frame trajectories, which were persisted only for those seeds, whereas the Skip-RNN comparison requires training an additional model on the same partitions. Both are controls on the mechanism rather than estimates of the main effect, and each states its seed range in the corresponding caption.

With $n = 12$ the exact paired Wilcoxon test resolves conventional significance: AF-TOI dominates the Baseline on F1 and ECE ($p = 0.0005$) and on $\mathrm{TTD}_{\mathrm{ef}}$ ($p = 0.021$), with AF-TOI attaining higher F1 in twelve of twelve paired seeds. For FR in the fixed

Table 8
Global factorial study results under continuous causal streaming (mean ± inter-seed standard deviation; twelve seeds, 42–53). Twelve-seed campaign executed with the same frozen pipeline as the initial four-seed campaign; ms/f is measured wall-clock time and is homogeneous within this campaign.

Model	MHEG	F1↑	ECE↓	FR↓	A_p (s)↑	TTD_b (s)↓	TTD_{ef} (s)↓	ms/f↓	%Sup↑
LSTM-Baseline	No	0.366 ± 0.069	0.686 ± 0.018	0.033 ± 0.054	0.657 ± 0.325	0.174 ± 0.114	0.500 ± 0.630	0.677 ± 0.373	0.000 ± 0.000
LSTM-Baseline	Yes	0.369 ± 0.072	0.685 ± 0.018	0.035 ± 0.053	0.468 ± 0.203	0.223 ± 0.145	0.570 ± 0.646	0.255 ± 0.135	60.996 ± 7.644
LSTM-AF-TOI	No	0.846 ± 0.180	0.495 ± 0.146	0.001 ± 0.005	1.200 ± 0.437	0.069 ± 0.106	0.083 ± 0.124	0.665 ± 0.338	0.000 ± 0.000
LSTM-AF-TOI	Yes	$\mathbf{0.846 \pm 0.180}$	$\mathbf{0.494 \pm 0.145}$	$\mathbf{0.001 \pm 0.005}$	$\mathbf{0.794 \pm 0.377}$	$\mathbf{0.085 \pm 0.124}$	$\mathbf{0.099 \pm 0.142}$	$\mathbf{0.340 \pm 0.167}$	$\mathbf{48.130 \pm 4.993}$

A_p: mean pre-onset anticipation in the progressive regime (higher is better; stable-detection search from frame 0; undetected episodes count as zero anticipation; not a component of TTD_b or TTD_{ef}). TTD_b: gross latency over detected episodes (partial, optimistic). TTD_{ef}: effective detection delay with miss penalty (primary arbiter). ECE: auxiliary indicator. %Sup: fraction of frames where MHEG suppressed computation (buffer reuse).

Table 9
Matched-pairs rank-biserial correlation (r_{rb}) and paired Wilcoxon test: Baseline Fixed vs. AF-TOI Fixed, $n = 12$ seeds.

Metric	r_{rb}↑	n_{eff}	p-value
F1	+1.000	12	0.0005
ECE	+1.000	12	0.0005
TTD_{ef}	+0.744	12	0.0210
FR	+1.000	5	0.0625

$n = 12$ seeds (42–53), exact two-sided paired Wilcoxon; r_{rb} is the matched-pairs rank-biserial correlation [50], oriented so that positive favors AF-TOI and computed on the n_{eff} nonzero paired differences (no verbal magnitude classes are attached). The FR p-value reflects ties at zero (seven of twelve Baseline seeds have FR = 0; the exact test discards the seven zero differences, leaving five informative pairs, so the FR row must be read jointly with its n_{eff}); in the scheduled contrast FR reaches $p = 0.0156$ ($r_{rb} = +1.000$, $n_{eff} = 7$).

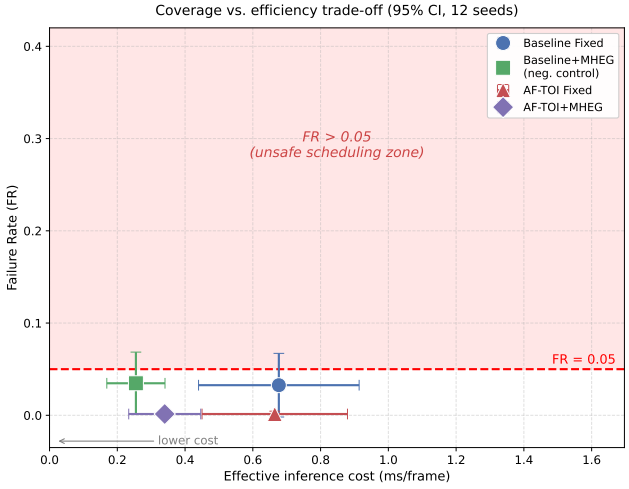


Figure 7: Global miss rate versus effective inference cost per frame (CI 95%, seeds 42–53). The dashed line marks the FR = 0.05 safety boundary; the shaded area denotes the unsafe scheduling region. The dashed line is the *aggregate* FR reference; trustworthiness is assessed per seed (Section 2.2) – Baseline+MHEG violates FR ≤ 0.05 in 3 of 12 seeds despite an aggregate mean below 0.05.

contrast the test yields $p = 0.0625$: seven of the twelve Baseline seeds also have FR = 0 and the exact two-sided signed-rank test discards zero differences, leaving five informative pairs, all favoring AF-TOI; only these five nonzero pairs contribute, and the contrast is not significant at the 0.05 level; in the scheduled contrast (Baseline+MHEG vs. AF-TOI+MHEG) FR reaches $p = 0.0156$. The negative control is likewise significant: adding MHEG to unorganized trajectories degrades TTD_{ef} ($p = 0.001$) without reducing FR, whereas on organized trajectories it reduces cost ($p = 0.0005$) while leaving the miss count of every seed unchanged. Direct operational means in the fixed contrast (AF-TOI Fixed vs. Baseline Fixed) match the four-seed campaign's pattern (FR 0.001 vs. 0.033; TTD_{ef} 0.083 s vs. 0.500 s). The complementary campaigns (post-hoc calibration, teacher ablation, stress scenarios, and gating baselines) use the initial four seeds (42–45) throughout. Table 9 summarizes.

Table 10 stratifies by operational regime, showing that the complete ecosystem preserves coverage in both progressive and abrupt profiles. Figure 7 synthesizes the trade-off. Baseline Fixed and Baseline+MHEG occupy nearly coincident FR positions (0.033 vs. 0.035), so selective computation without temporal organization gives no coverage benefit; AF-TOI+MHEG is the only configuration reaching low cost and FR below the safety threshold at once.

4.5. Movement 4. Distributional Structure

Mean TTD_{ef} can mask heterogeneous per-episode behavior. Figures 8 and 9 show the pattern: AF-TOI histograms concentrate near zero, while variants without organization spread widely, with 47 episodes stuck at the 10 s miss penalty, precisely the episodes where the scheduler failed to invest at emergence windows.

The high standard deviations of the non-AF-TOI configurations are not stochastic noise but the 47 severely penalized episodes where computation was suppressed at critical windows. With AF-TOI, the miss population collapses from 47 to 2 (out of 1,440 critical test episodes), so the induction restructures the failure distribution rather than merely lowering mean

Table 10
Performance stratified by operational regime. TTD_a denotes gross detection latency within the abrupt regime (detected episodes only), measured from the onset like TTD_b; A_p denotes mean pre-onset anticipation in the progressive regime (higher is better; not a latency) (mean ± sd inter-seed, seeds 42–53).

	Progressive Regime (60%)		Abrupt Regime (40%)		Global	
Configuration	**FR**	**A_p (s)**	**FR**	**TTD_a (s)**	**FR**	**$\mathrm{TTD}_{\mathrm{ef}}$ (s)**
Baseline Fixed	0.000 ± 0.000	0.657 ± 0.325	0.082 ± 0.136	0.467 ± 0.357	0.033 ± 0.054	0.500 ± 0.630
Baseline + MHEG	0.002 ± 0.005	0.468 ± 0.203	0.083 ± 0.135	0.492 ± 0.379	0.035 ± 0.053	0.570 ± 0.646
AF-TOI Fixed	0.000 ± 0.000	1.200 ± 0.437	0.004 ± 0.012	0.123 ± 0.264	0.001 ± 0.005	0.083 ± 0.124
AF-TOI + MHEG	0.000 ± 0.000	0.794 ± 0.377	0.004 ± 0.012	0.141 ± 0.284	0.001 ± 0.005	0.099 ± 0.142

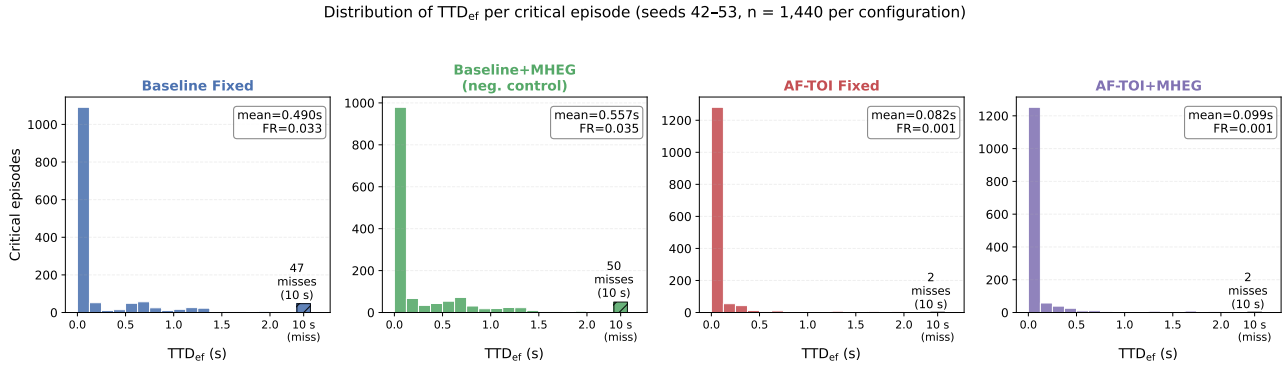


Figure 8: $\mathrm{TTD}_{\mathrm{ef}}$ frequency distributions per critical episode across seeds 42–53 ($n = 1{,}440$ per configuration). The 10 s value denotes the miss penalty used in $\mathrm{TTD}_{\mathrm{ef}}$. The in-panel mean is the per-episode average (detected delay, or 10 s when missed) and therefore differs slightly from the seed-level $\mathrm{TTD}_{\mathrm{ef}}$ of Table 8, which averages the detected delay and adds $\mathrm{FR} \times T_{\mathrm{ep}}$ at seed level.

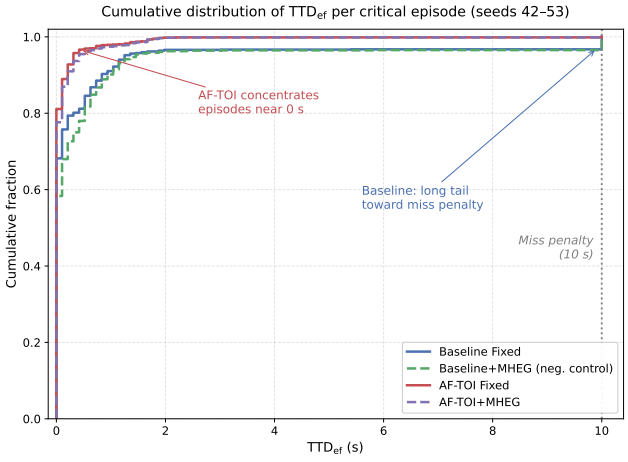


Figure 9: Cumulative distribution functions of $\mathrm{TTD}_{\mathrm{ef}}$ per critical episode across seeds 42–53 ($n = 1{,}440$ per configuration; twelve-seed campaign).

latency. Baseline Fixed and Baseline+MHEG share nearly identical miss counts (47 vs. 50), indicating no distributional coverage benefit from selective computation without organization. Adding MHEG to already organized trajectories does not shift the profile, since the scheduler suppresses genuinely redundant intervals without creating new failures.

4.6. Movement 5. Robustness

Figure 10 shows performance across the twelve seeds with 95% CI. AF-TOI+MHEG stays below FR = 0.05 in every run with lower variability than the other configurations, while variants without organization cross the coverage threshold in at least one seed. Trustworthy selective computation thus requires the enabling property to be present consistently.

Table 11 separates a *mechanism-removal ablation* from *parameter sensitivity*, over the twelve seeds of the factorial campaign and with exact values. All five conditions were retrained from scratch and evaluated under a single frozen replication protocol (the replay pipeline that also underlies Sections 4.10 and 4.11); because that protocol differs from the main-campaign pipeline of Table 8, the Full AF-TOI row serves as the within-table reference and its values are not numerically comparable to Table 8. The no-temporal-organization condition removes only the onset-anchored terms of the AF-TOI loss (the onset-window tracking penalties, the post-onset asymmetric penalty, and the pre-onset push), preserving the reference model, the alignment

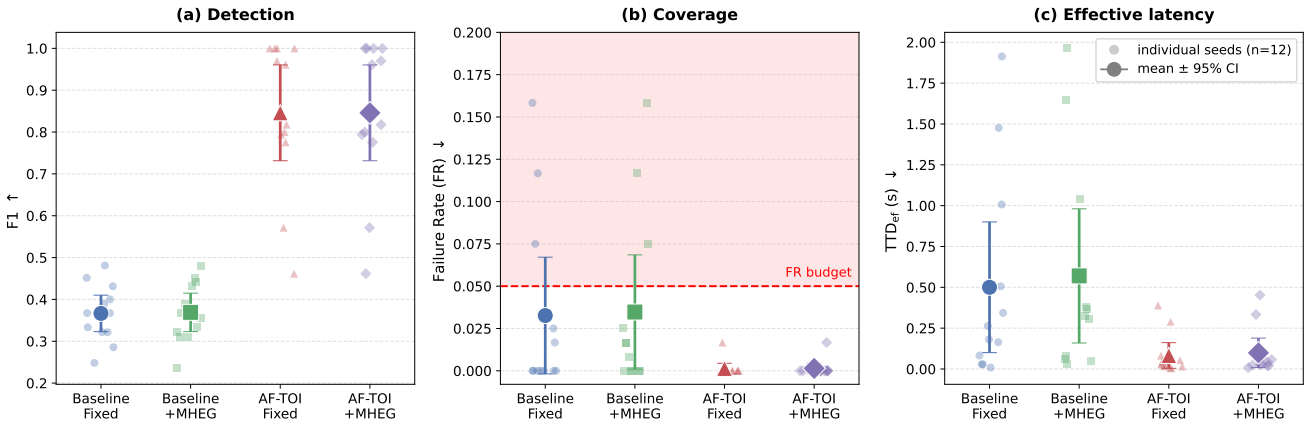


Figure 10: Per-seed performance across twelve independent partitions (CI 95%, seeds 42–53). The dashed line marks the FR = 0.05 safety boundary.

loss, the architecture, the data, and the calibration protocol.

Removing temporal organization leaves F1 and raw coverage at their ceiling but is consistently worse on the scheduled contrast, which is the configuration this paper deploys: under MHEG, its $\mathrm{TTD}_{\mathrm{ef}}$ is ≈3.5× the canonical value (0.180 vs. 0.051 s; exact paired Wilcoxon $p = 0.005$, matched-pairs rank-biserial $r_{\mathrm{rb}} = 0.87$ against removal), ≈3× the *worst* value in the α grid (0.059 s at $\alpha = 0.05$; $p = 0.003$, $r_{\mathrm{rb}} = 0.90$), and ≈4.8× the grid mean ($p = 0.001$). Episode-level ECE ($\mathrm{ECE}_{\mathrm{v29}}$) degrades against every grid point in twelve of twelve seeds (0.642 vs. 0.528–0.535; $p = 0.0005$), and the ablated model suppresses *more* (59.0% vs. 47.5%; $p = 0.021$) while detecting later, so the additional suppression is not safe allocation.

Under the fixed policy the contrast is significant against the canonical $\alpha = 0.15$ (≈5.7×, 0.250 vs. 0.043 s; $p = 0.016$, $r_{\mathrm{rb}} = 0.77$) but not against the worst grid point (0.161 s at $\alpha = 0.30$; $p = 0.91$). Sensitivity to α is therefore not negligible under the fixed policy, where $\mathrm{TTD}_{\mathrm{ef}}$ spans 0.043–0.161 s across the grid; the mechanism-versus-tuning distinction rests on the scheduled contrast and on calibration, where removal is worse than every evaluated α.

Table 12 checks robustness to the miss penalty. The relative ordering across configurations holds for 5, 10, and 15 s penalties, so the complete-ecosystem advantage does not depend on the specific T_{ep}.

4.7. Movement 6. Emergent Organization

Movement 6 examines a property not directly optimized: whether AF-TOI organization extends to intermediate risk, producing outputs coherent with SHRP2 gradients. In borderline scenarios unseen during training, the Attention regime records $\bar{p}_{\max} = 0.048 \pm 0.195$ (5.8% activations above θ; $\bar{p}_{\max}$ denotes the per-episode maximum of the emitted frame probability p_t, averaged over the episodes of a severity level – a signal-level descriptor, distinct from the stable-detection rule that defines FR in Table 2) and Alert $\bar{p}_{\max} = 0.380 \pm 0.416$ (48.8%), evaluated on seed 42; on the factorial test partition of the same seed, Normal episodes stay at $\bar{p}_{\max} = 0.000$ (0.0%) and Critical episodes reach 0.940± 0.083 (100.0%). Table 13 and Figure 11 summarize.

The results give preliminary, descriptive evidence that the induction extends beyond the training recipes. Normal and Attention episodes stay mostly below the activation threshold, while Alert and Critical trigger activation with progressively higher frequency, qualitatively aligned with increasing risk exposure despite no explicit severity objective during training.

Both summaries order the four levels monotonically ($\bar{p}_{\max}$: $0.000 < 0.048 < 0.380 < 0.940$; activation fraction: $0.0 < 5.8 < 48.8 < 100.0\%$). Two cautions bound this reading: the Attention and Alert rows come from a different checkpoint and protocol than the Normal and Critical rows (Table 13 note), so the cross-level ordering is qualitative rather than a single-model measurement; and the large variance within Attention and Alert (episodes split between near-zero and high-response trajectories) makes the activation fraction the more stable summary within those levels.

The evidence therefore does not support the claim that AF-TOI learned a severity scale, and – because the four levels are not all measured under one checkpoint and protocol – the joint ordering is a description across two evaluation pipelines, not a single-model measurement. Within that limit, the observed pattern is a progressively higher activation fraction at higher severity, including for intermediate-risk conditions unseen in training. Whether this holds on naturalistic data, and whether full ordinal separation can emerge under broader conditions, remain open (Section 5).

Table 11
Mechanism-removal ablation and α sensitivity (twelve seeds 42–53, mean ± sd). All five conditions were retrained and evaluated under the same frozen replication protocol (the replay pipeline of Sections 4.10 and 4.11), which differs from the main-campaign pipeline of Table 8; values are comparable within this table only. F1 and ECE use the fixed policy. The no-temporal condition zeroes only the onset-anchored loss terms. Pairwise contrasts in the text are exact two-sided paired Wilcoxon tests over the twelve seeds with matched-pairs rank-biserial effect sizes, uncorrected for multiplicity and classified as secondary under the taxonomy of Section 4.1.

			Fixed policy		MHEG policy		
Condition	**F1↑**	**ECE↓**	**FR↓**	**$\mathrm{TTD}_{\mathrm{ef}}$ (s)↓**	**FR↓**	**$\mathrm{TTD}_{\mathrm{ef}}$ (s)↓**	**%Sup↑**
No temporal organization	0.884 ± 0.022	0.642 ± 0.012	0.013 ± 0.046	0.250 ± 0.458	0.001 ± 0.002	0.180 ± 0.104	59.0 ± 9.3
$\alpha = 0.05$	0.902 ± 0.034	0.528 ± 0.019	0.008 ± 0.016	0.112 ± 0.160	0.000 ± 0.000	0.059 ± 0.091	50.1 ± 9.5
Full AF-TOI ($\alpha = 0.15$)	0.895 ± 0.032	0.531 ± 0.017	0.002 ± 0.007	$\mathbf{0.043 \pm 0.082}$	0.000 ± 0.000	0.051 ± 0.097	47.5 ± 10.0
$\alpha = 0.30$	0.885 ± 0.017	0.533 ± 0.016	0.015 ± 0.025	0.161 ± 0.257	0.001 ± 0.005	0.030 ± 0.060	48.6 ± 9.3
$\alpha = 0.50$	0.892 ± 0.024	0.535 ± 0.019	0.008 ± 0.017	0.079 ± 0.172	0.001 ± 0.002	0.009 ± 0.027	50.0 ± 7.2

Table 12
$\mathrm{TTD}_{\mathrm{ef}}$ sensitivity as a function of miss penalty; values recomputed for the twelve-seed campaign.

Configuration	Penalty 5.0 s	Penalty 10.0 s	Penalty 15.0 s
Baseline Fixed	0.337	0.500	0.663
Baseline+MHEG	0.396	0.570	0.743
AF-TOI Fixed	0.076	0.083	0.090
AF-TOI+MHEG	0.092	0.099	0.106

Computed analytically from $\mathrm{TTD}_{\mathrm{ef}} = \mathrm{TTD}_b + \mathrm{FR} \times T_{\mathrm{ep}}$.

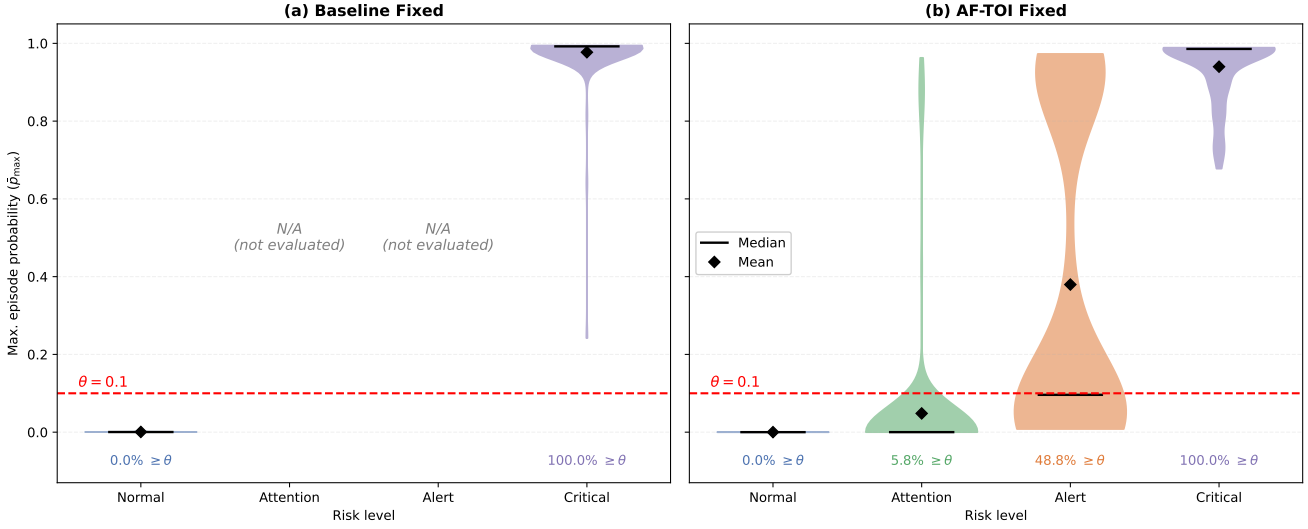


Figure 11: $\bar{p}_{\max}$ distributions by severity level in borderline episodes for seed 42. Baseline Fixed contains Normal and Critical levels; AF-TOI Fixed includes Normal, Attention, Alert, and Critical levels with activation fractions reported in the plot. The Normal and Critical distributions are drawn from the same main-test-set data as Table 13, ensuring figure–table consistency.

4.8. Post-hoc Calibration Control

A complementary question is whether temporal organization could be approximated by post-hoc calibration of Baseline probabilities. Three techniques were applied: temperature scaling (TS) [12], Platt scaling (PS) [39], and isotonic calibration (IC) [40]. MHEG was applied to each variant with thresholds re-calibrated per seed to maximize suppression under FR $\leq$ 0.05, yielding Table 14.

Probabilistic alignment improves in all variants. Uncalibrated Baseline ECE is $\approx$ 0.035; after isotonic

Table 13
Probabilistic response in borderline episodes (seed 42, AF-TOI Fixed configuration).

Risk Level	Behavior	$\bar{p}_{\max}$	Fraction
Normal	Stable baseline driving	0.000 ± 0.000	0.0%
Attention	Environmental perturbation	0.048 ± 0.195	5.8%
Alert	Mild distraction at low speed	0.380 ± 0.416	48.8%
Critical	Severe distraction with imminent risk	0.940 ± 0.083	100.0%

Fraction: proportion of episodes with $\bar{p}_{\max} \geq \theta = 0.10$. Normal and Critical rows: factorial test partition of seed 42 under the prefix-window inference protocol of Section 4.1 (the same probability stream behind Table 8); Attention and Alert rows: independently generated borderline recipes evaluated under the replication-protocol model (Section 4.10 pipeline), so the cross-level comparison is qualitative.

calibration it drops to 0.0075 (reduction 0.0274). The techniques work as intended.

A note on scale: ECE values in this paper are protocol-specific and are therefore named: $\mathrm{ECE}_{\mathrm{pipe}}$ (the factorial evaluation of Table 8; episode-level – episode scores against episode labels – under the main-campaign pipeline), $\mathrm{ECE}_{\mathrm{replay}}$ (the frame-level reliability diagram of Figure 4, replay logs of seed 42), $\mathrm{ECE}_{\mathrm{cal}}$ (this subsection), and $\mathrm{ECE}_{\mathrm{v29}}$ (the mechanism-removal ablation of Table 11; episode-level, replication protocol). Each computes ECE over a different probability stream, partition, and binning; magnitudes are comparable within, but not across, these analyses (Table 23). All four variants use ten equal-width bins over the emitted probabilities; $\mathrm{ECE}_{\mathrm{replay}}$ additionally discards bins with fewer than ten frames (bin weights keep the full-sample denominator, so discarded bins contribute zero).

Their operational profiles under MHEG differ sharply. Temperature scaling degrades $\mathrm{TTD}_{\mathrm{ef}}$ from 0.2324 s to 0.4377 s, reproducing the failure pattern of the negative control. Platt scaling and isotonic calibration reach lower $\mathrm{TTD}_{\mathrm{ef}}$ under MHEG (0.1221 s and 0.0527 s), which needs a precise mechanistic reading to avoid misinterpretation.

Post-hoc calibration is a monotonic transformation of the output probabilities: it preserves their ordering along time, though not necessarily the ordering of the binary entropies. Whether the temporal structure of $H(t)$ changes under a given calibrator is therefore measured here rather than assumed. Figure 12 is consistent with this. In panel (a), TS, PS, and IC preserve the Baseline temporal morphology, and the entropy profile around onset stays undifferentiated. In panel (b), AF-TOI structurally shifts the trajectory, elevating $H(t)$ near onset and suppressing it in stable phases.

The lower $\mathrm{TTD}_{\mathrm{ef}}$ for PS and IC under MHEG does not come from temporal reorganization. Both redistribute the probability scale, and IC in particular compresses probabilities toward empirical frequencies, shifting the operating regime of the entropy-based gate; in the measured trajectories (Figure 12a) the temporal entropy profile around onset is unchanged. When MHEG thresholds are re-calibrated to that scale, lower suppression cost is achievable at a given coverage. This is a threshold calibration effect, not an organizational one: the calibrated operating points in Figure 13(a) stay within the calibration family's own region of its $\mathrm{ECE}_{\mathrm{cal}}$ panel. AF-TOI, by contrast, induces an organization regime that probability-scale adjustment does not reproduce, visible in the temporal morphology of panel (b): $H(t)$ is elevated around the onset and suppressed in stable phases, a differentiation that tracks onset proximity and is absent from every calibrated variant in panel (a).

A scalar summary, the entropy-stratification score $H_{\mathrm{strat}} = (H_{\mathrm{post}} - H_{\mathrm{stable}})/H_{\max}$, is not reported: under a single unified procedure on the stored trajectories it is sensitive to the probability scale rather than to temporal organization (the calibrated variants score 0.63–0.82, above both the uncalibrated Baseline, 0.39 ± 0.06, and AF-TOI, 0.34 ± 0.03), so it cannot support – in either direction – claims about organization. The evidence for this control therefore rests on the trajectory morphology (Figure 12), the operational locus (Figure 13), and Table 14.

Table 14 also contains the strongest calibrated configuration: Baseline+IC+MHEG reaches $\mathrm{TTD}_{\mathrm{ef}} = 0.0527$ s with FR = 0.0000 at $\approx 57\%$ suppression, which matches or exceeds AF-TOI+MHEG on the endpoints of this in-distribution replay. The distinction is mechanistic, not an endpoint ranking on this replay: the isotonic gains arise from threshold recalibration on a frozen probability scale, remain tied to the distribution on which the calibrator was fitted, and are reported on four seeds without inferential statistics, whereas the organized signal is what the stress campaign of Section 4.10 and the temporal-scale campaign of Section 4.12 probe under controlled shift. Whether the isotonic advantage survives such shifts has not been tested here; it remains an open control, and this is a limitation of the present comparison.

Figure 13 shows the measured operating points against $\mathrm{TTD}_{\mathrm{ef}}$ (point estimates with 95% CIs, seeds 42–45), in two panels because the two ECE variants are not comparable (Table 23): panel (a) places the calibration family on its own $\mathrm{ECE}_{\mathrm{cal}}$ axis, where Platt and isotonic variants reach low ECE and, combined with MHEG, low $\mathrm{TTD}_{\mathrm{ef}}$; panel (b) places the four reference configurations on the episode-level $\mathrm{ECE}_{\mathrm{pipe}}$ axis, where the AF-TOI configurations attain the lowest $\mathrm{TTD}_{\mathrm{ef}}$. No cross-panel ECE comparison is implied; the panels display measured points, not frontiers.

Probability-scale correction does not modify the internal representation or how it responds to risk proximity; it rescales declared confidence at the output without inducing the temporal differentiation reliable

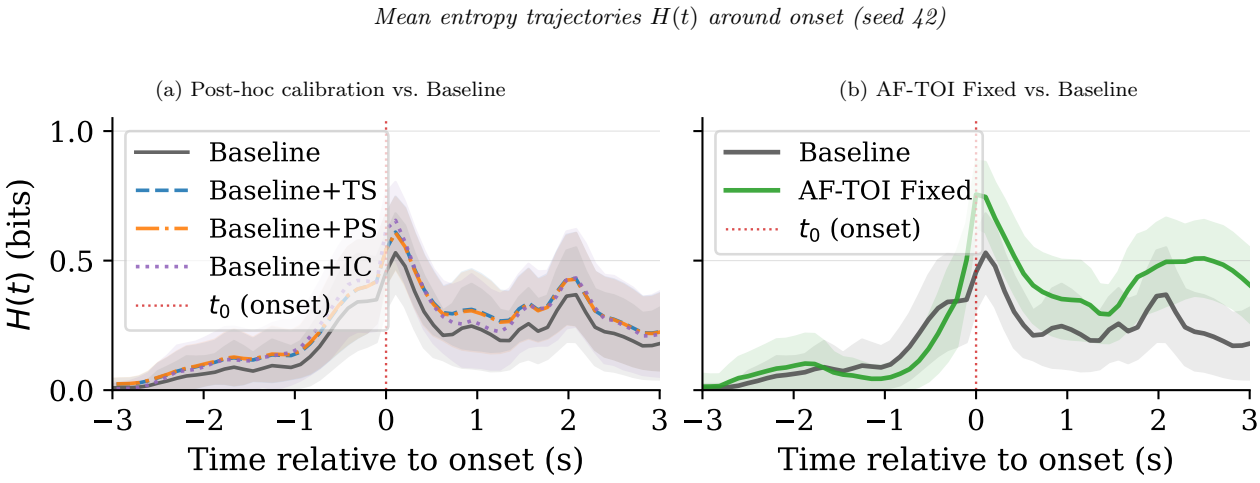


Figure 12: Mean entropy trajectory $H(t)$ around onset for seed 42; aggregate H_{pre} metrics are consistent across seeds 42–45. Panel (a) shows post-hoc calibration variants; panel (b) shows AF-TOI.

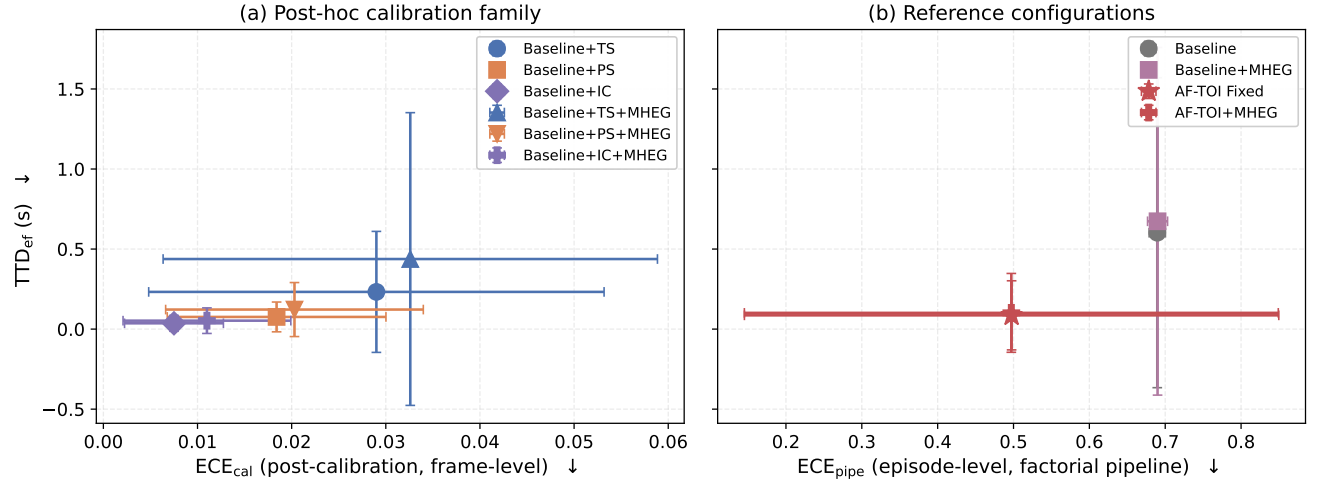


Figure 13: Operating points vs. TTD$_{\text{ef}}$ (CI 95%, seeds 42–45). Panel (a): post-hoc calibration family on the frame-level ECE$_{\text{cal}}$ axis; panel (b): reference configurations on the episode-level ECE$_{\text{pipe}}$ axis. The two ECE variants are protocol-specific and not comparable, hence the separate axes (Table 23).

scheduling needs. For the reason given above, no per-seed stratification-score figure is reported.

The two routes to low TTD$_{\text{ef}}$ differ in mechanism. Reaching it through threshold recalibration to a redistributed probability scale differs from reaching it through induced temporal differentiation. In the former, the gain is contingent on the calibrator's particular distribution and may not survive distributional shift, since the gate threshold has no fixed relation to the transition dynamics it must protect. In the latter, the scheduler exploits a structural trajectory property deliberately induced during training and tied to the causal dynamics of the process. The trajectory morphology of Figure 12 and the measured operating points in Figure 13 indicate that the two regimes are empirically distinguishable in the measured trajectories and operating points.

4.9. Architectural Ablation of the Non-Causal Reference

To test whether the organization gains depend on the specific BiLSTM topology, an ablation used a simplified protocol (60 epochs, no three-phase scheduler).

Two deliberate simplifications explain why the absolute values in Table 15 differ from the main study and must not be compared directly. First, the three-phase alignment scheduler, a constitutive element of AF-TOI that stabilizes coupling and improves induced

Table 14
Epistemological control via post-hoc calibration applied to Baseline with and without MHEG. MHEG thresholds are re-calibrated per seed for each variant. ECE (cal): post-calibration ECE; ΔECE: improvement vs. uncalibrated Baseline (≈ 0.035). Reference(twelve-seed campaign): Baseline Fixed FR = 0.033 ± 0.054, $\text{TTD}_{\text{ef}} = 0.500 \pm 0.630$ s; AF-TOI+MHEG FR = 0.001 ± 0.005, $\text{TTD}_{\text{ef}} = 0.099 \pm 0.142$ s, 48.1% computation suppression.

Configuration	ECE (cal)↓	ΔECE↑	F1↑	FR↓	TTD_b (s)↓	TTD_{ef} (s)↓	%Sup↑
Baseline+TS	0.0290	0.0059	0.4410	0.0104	0.1283	0.2324	0.0%
Baseline+TS+MHEG	0.0326	0.0062	0.4835	0.0250	0.1877	0.4377	58.8450%
Baseline+PS	0.0184	0.0165	0.4687	0.0000	0.0760	0.0760	0.0%
Baseline+PS+MHEG	0.0203	0.0170	0.4608	0.0021	0.1013	0.1221	62.0800%
Baseline+IC	0.0075	0.0274	0.5042	0.0000	0.0362	0.0362	0.0%
Baseline+IC+MHEG	0.0110	0.0254	0.5040	0.0000	0.0527	0.0527	57.1100%

TTD_b: gross latency over detected episodes (partial). TTD_{ef}: effective detection delay with miss penalty. ΔECE: positive indicates improvement. %Sup: computation suppression rate by MHEG. Lower TTD_{ef} for PS and IC variants reflects threshold recalibration to the redistributed probability scale, not temporal trajectory reorganization (see the entropy trajectories of Figure 12; no scalar stratification score is reported, see text).

Table 15
Non-causal reference architectural ablation (simplified protocol, 60 epochs, no 3-phase scheduler). Values not comparable with main study.

Causal Target Arch.	Non-Causal Reference Arch.	Approach	FR↓	TTD_{ef} (s)↓
Causal LSTM	None	Baseline Fixed	0.713 ± 0.069	7.223 ± 0.763
Causal LSTM	**BiLSTM arch.**	**AF-TOI method**	**0.275 ± 0.189**	**3.089 ± 1.837**
Causal LSTM	Transformer	TransTOI method	0.169 ± 0.163	2.022 ± 1.860
Causal Transformer	None	Baseline Fixed	0.958 ± 0.030	—
Causal Transformer	BiLSTM	AF-TOI method	0.467 ± 0.414	—

Absolute FR values exceed the main factorial study because the scheduler was omitted to isolate the reference architecture effect. TTD_{ef} omitted for Causal Transformer configurations due to near-complete coverage failure (FR ≈ 0.96), rendering the metric uninformative. This reinforces the scheduler's importance for transferred organization quality, without invalidating the non-causal bidirectional structural signal principle.

organization, was omitted. Second, MHEG was excluded to isolate reference topology, which removes the coverage-preserving layer and drives FR to deployment-unacceptable levels. The question is strictly comparative: which reference topology produces greater relative organization in the causal target under otherwise identical conditions.

With the causal LSTM target, TransTOI produces organization comparable to AF-TOI, consistent with induction deriving from the non-causal bidirectional structural signal rather than the specific BiLSTM topology. The causal Transformer target collapses in coverage without external alignment, supporting the causal LSTM as the most robust architecture for embedded deployment.

4.10. Negative Controls and Cost-Matched Robustness

To probe the central claim beyond the 2×2 factorial, we ran a complementary controlled campaign over nine onset/noise scenarios (Table 16), each with 360 episodes across seeds 42–45. The protocol adds two instruments not used in the main study. First, a *compute-rate-matched* budget forces every negative control to spend the same fraction of frames on recurrent updates as MHEG (±2%), so that any difference reflects *where* computation is spent, not *how much*. Second, a near-onset false-suppression metric (*FS@onset*, the fraction of onset-window frames whose update was suppressed) isolates the specific failure that unsafe scheduling produces. Coverage and miss rate are reported as percentages; compute rate is the fraction of frames on which the update runs. These magnitudes are on the stress-protocol scale and are not comparable to the aggregate TTD_{ef} of Section 4.4; the study is designed to rank safety at fixed cost, not to re-estimate efficiency.

The budget-matched controls used here (random, unstructured, and lagged allocators) are intentionally harsher than the Baseline+MHEG negative control of Section 4.4: they hold the compute budget fixed and strip temporal structure, isolating the effect of *where* computation is placed rather than how much.

Cost-matched safety. The most informative result of this campaign is that, under its stress protocol and at matched budget, structure rather than compute budget drives near-onset safety; Section 4.12 qualifies the scope of this reading under temporal-scale shift, where the

Table 16
Controlled robustness scenarios (each 360 episodes, seeds 42–45).

Scenario	Onset	Entropy	Stressed failure mode
balanced_onsets	mixed	org.	cost saving (reference)
abrupt_onset_dominant	abrupt	org.	false suppression at abrupt onset
progressive_onset_dominant	progressive	org.	delayed detection ($\mathrm{TTD}_{\mathrm{ef}}$)
rare_abrupt_critical	abrupt	org.	missed transient event (coverage)
high_noise_entropy	mixed	degr.	entropy scheduler misfires
kinematic_signal_missing	abrupt	org.	loss of kinematic protection
entropy_lag	abrupt	degr.	lagged entropy misses onset
false_onset	mixed	org.	wasted compute / false trigger
no_onset_negative_control	mixed	org.	sanity: FS must be ≈ 0

budget-matched advantage concentrates on coverage and timeliness rather than on suppression placement. Table 17 matches every control to the MHEG budget within ±2%. At *identical* cost, MHEG preserves full coverage with zero onset-window suppression, while budget-matched controls degrade near onset: under abrupt onsets the random and unstructured controls reach 41–44% FS@onset, and under rare transient events coverage collapses to 22–24% (miss rate $\geq 75\%$) at the same budget. Spending the same computation without temporal structure is therefore unsafe.

Negative-control sweep. Across all nine scenarios, MHEG (hybrid and calibrated) held FS@onset at 0.0 wherever the kinematic channel was available, whereas the budget-matched unstructured (NC2/NC6), random (NC1), and lagged (NC3) controls incurred near-onset suppression on every abrupt-dominated scenario, peaking at 56.7% FS@onset for the lagged control under abrupt onsets. The `no_onset_negative_control` sanity check behaved as required, with false suppression ≈ 0 for all methods. When the kinematic channel was removed (`kinematic_signal_missing`), MHEG retained coverage 100% with only 11.5% FS@onset, while the matched controls became unsafe (33–46% FS@onset), a result consistent with the kinematic trigger acting as the intended fallback for sub-onset windows.

Entropy-lag sensitivity. Table 18 varies the entropic-signal lag. Progressive onsets tolerate lag (coverage stays 100% through 10 frames), but abrupt and especially rare transient onsets degrade sharply: at a 5-frame lag the rare-transient scenario loses all coverage (miss 100%). This is the mechanistic signature of the paper's thesis, namely that near-onset entropy timing, not aggregate calibration, governs safe suppression.

Ablation of the scheduling channels. A per-channel ablation indicates that both components are necessary. Entropy-only scheduling preserves coverage but adds delay ($\mathrm{TTD}_{\mathrm{ef}}$ rises relative to the hybrid), while kinematic-only scheduling is catastrophic on progressive onsets, collapsing coverage to 0% because slow transitions never trigger the kinematic gate. The hybrid matches the always-compute delay at a fraction of the cost, and paired tests over seeds rank the hybrid strictly above the entropy-only control on false suppression with no delay penalty against always-compute (no paired difference).

4.11. Confidence-Threshold and Smoothed-Entropy Baselines

Two additional scheduling baselines complete the comparison requested by the selective-inference literature: a *confidence-threshold gate*, which updates only when the buffered confidence is low (equivalently, when $H(p_{t-1}) \geq \tau$), and a *smoothed-entropy gate*, which thresholds a moving average of the buffered entropy (window w=5). Both use no kinematic channel. They are simulated under a replay protocol on the stored full-inference trajectories of the factorial study (seeds 42–45): on update, the buffered probability is refreshed from the stored trajectory; on suppression, it is held. Detection latencies are measured from the label onset (the first frame with $y > 0.35$), and the kinematic trigger of the reference gate uses the monitored kinematic channels of Eq. 10. Thresholds are oracle-calibrated on the test partition (maximum suppression subject to FR ≤ 0.05, jointly with the detection threshold), an upper bound that *favors* the baselines. Magnitudes are on the replay scale and comparable within Table 19 only.

Table 19 reports two operating points. At the safe point, the purely entropic gates collapse: without an external wake-up signal, a buffered low-entropy value never fires again (the gate deadlocks), so any non-negligible suppression forfeits coverage, and the maximum safe suppression is 0–2% on either trajectory type. Temporal smoothing does not repair this, since averaging a frozen buffer changes nothing. At the matched-suppression point (tuned to spend the same fraction of frames as the entropy+kinematic gate), the confidence and smoothed gates miss 99% of critical episodes on unorganized trajectories and 59% on organized ones, whereas the entropy+kinematic gate sustains mean FR ≤ 0.010 at 93.6% and 69.3% suppression. This table therefore isolates one claim: the

Table 17
Cost-matched comparison (±2% compute budget). MHEG vs. budget-matched negative controls. Coverage, miss and FS@onset in percent; compute rate is the fraction of frames updated.

Scenario	Method	Compute rate	Coverage	Miss	FS@onset
balanced_onsets	**MHEG (hybrid)**	0.217	**100.0**	**0.0**	**0.0**
	NC1 random budget-matched	0.218	99.7	0.3	23.9
	NC2 unstructured aggressive	0.218	100.0	0.0	20.3
	NC6 cost-matched unstructured	0.218	100.0	0.0	20.3
abrupt_onset_dominant	**MHEG (hybrid)**	0.193	**100.0**	**0.0**	**0.0**
	NC1 random budget-matched	0.195	99.7	0.3	43.9
	NC2 unstructured aggressive	0.196	99.7	0.3	41.6
	NC6 cost-matched unstructured	0.196	99.7	0.3	41.6
rare_abrupt_critical	**MHEG (hybrid)**	0.066	**100.0**	**0.0**	**0.0**
	NC1 random budget-matched	0.066	21.9	78.1	86.2
	NC2 unstructured aggressive	0.067	24.4	75.6	85.1
	NC6 cost-matched unstructured	0.067	24.4	75.6	85.1

Compute rate is the per-scenario fraction of frames updated under the matched-budget protocol; it is deliberately distinct from, and not comparable to, the aggregate suppression rate (%Sup) of Section 4.4. Within each scenario the budget is matched *across methods* to isolate allocation quality at equal cost.

Table 18
Entropy-lag sensitivity by onset type. Coverage, miss and FS@onset in percent.

Lag	Scenario	Coverage	Miss	FS@onset
1	abrupt_onset_dominant	100.0	0.0	28.3
1	rare_abrupt_critical	100.0	0.0	26.3
3	abrupt_onset_dominant	100.0	0.0	56.7
3	rare_abrupt_critical	64.4	35.6	78.8
5	abrupt_onset_dominant	100.0	0.0	85.0
5	rare_abrupt_critical	0.0	100.0	100.0
10	rare_abrupt_critical	0.0	100.0	100.0

unconditional kinematic trigger is a structural necessity for any buffered entropy gate (Section 3.3). On this in-distribution replay the kinematic trigger alone keeps both trajectory types safe, so the specific contribution of temporal organization is not visible here; it is isolated by the mechanism-removal ablation of Table 11, by the `kinematic_signal_missing` scenario of Section 4.10, in which only the entropic component remains protective, and by the temporal-scale campaign of Section 4.12. The organized trajectories suppress less under the hybrid gate (69.3% vs. 93.6%) because their entropic component fires around transitions, which is its intended function.

A natural objection is that a deployed confidence gate would include a periodic refresh rather than trust a frozen buffer indefinitely. We therefore also evaluated a periodic-refresh confidence gate (forced update every R frames, otherwise update on high buffered entropy) under the same replay protocol (seeds 42–45, same oracle calibration on the test partition as Table 19; reported descriptively). The refresh breaks the deadlock by construction: at R=8, Baseline trajectories reach 85.9± 0.4% suppression with FR = 0 and $\mathrm{TTD}_{\mathrm{ef}}$ = 0.161 ± 0.063 s, and organized trajectories reach 59.9 ± 1.8% at 0.129 ± 0.076 s. On this in-distribution replay a blind cadence therefore suffices for the detection endpoints – consistent with the Skip-RNN result below, where endpoints likewise do not separate the designs. What the cadence does not provide is temporal allocation: a forced refresh belongs to the structure-free, budget-matched allocation family stressed in Section 4.10, whose random and unstructured members incur 41–44% FS@onset at equal compute under abrupt onsets and collapse to 22–24% coverage on rare transient events; the periodic-refresh gate itself was not rerun under that stress protocol, so this is a family-level reading rather than a direct measurement. The Table 19 deadlock should accordingly be read as a property of purely buffered confidence gates, not as a claim that no confidence-based scheduler can work.

A learned adaptive temporal-computation baseline (Skip-RNN) To complement these controls with a published method that allocates computation on the same axis, we trained Skip-RNN [6] end-to-end on the same data and partitions (seeds 42–45): a two-layer causal LSTM of the deployed size whose learned binary gate skips recurrent state updates, with the update budget controlled by the cost weight of the original formulation. Skip-RNN is not an uncertainty-aware method; it is reported here as a published adaptive-computation control. All systems in Table 20 are evaluated under a single harmonized protocol (detection from the label onset, m=3 consecutive frames at θ = 0.10, $\mathrm{TTD}_{\mathrm{ef}}$ as in Section 4.1), alongside the temporal semantics of the emitted signal: the fraction of pre-onset frames of critical episodes already above θ (PreOcc) and the fraction of frames of normal episodes above θ (NormOcc). The matched-suppression point equalizes the

Table 19
Confidence-threshold and smoothed-entropy baselines under the replay protocol (mean ± sd, seeds 42–45). Safe point: maximum suppression subject to FR ≤ 0.05. Matched point: thresholds tuned to the suppression level of the entropy+kinematic gate on the same trajectories. Oracle calibration on the test partition favors the baselines. Latencies are measured from the label onset; the kinematic trigger uses the monitored channels of Eq. 10.

		Safe point (FR ≤ 0.05)		Matched-suppression point		
Trajectories	**Gate**	**%Sup↑**	**TTD_{ef} (s)↓**	**%Sup**	**FR↓**	**TTD_{ef} (s)↓**
Baseline (unorg.)	Confidence-threshold	0.4 ± 0.7	0.01 ± 0.01	97.1 ± 0.2	0.990 ± 0.004	9.90 ± 0.04
	Smoothed entropy (w=5)	2.1 ± 3.6	0.11 ± 0.20	96.7 ± 0.5	0.988 ± 0.005	9.88 ± 0.05
	Entropy + kinematic	$\mathbf{93.6 \pm 0.0}$	0.25 ± 0.08	—	$\mathbf{0.002 \pm 0.004}$	—
AF-TOI (org.)	Confidence-threshold	0.0 ± 0.0	0.04 ± 0.04	69.7 ± 1.8	0.594 ± 0.128	5.94 ± 1.28
	Smoothed entropy (w=5)	0.0 ± 0.0	0.04 ± 0.04	69.7 ± 1.8	0.594 ± 0.128	5.94 ± 1.28
	Entropy + kinematic	$\mathbf{69.3 \pm 1.7}$	$\mathbf{0.35 \pm 0.18}$	—	$\mathbf{0.010 \pm 0.008}$	—

The entropy+kinematic row is the reference gate under the same replay protocol; its matched point coincides with its safe point. Confidence and smoothed gates coincide at the matched point because smoothing a frozen buffer does not alter the deadlock. On AF-TOI trajectories both purely entropic gates reach zero suppression at the safe point, so their safe-point rows coincide as well.

fraction of executed updates, not compute: an executed Skip-RNN update costs one recurrent step, whereas each executed update of the frozen AF-TOI(+MHEG) replay recomputes a W-frame window (Section 3.4), so equal suppression does not imply equal MACs.

At matched suppression (50.9% vs. 48.7%), Skip-RNN matches or exceeds the frozen AF-TOI(+MHEG) models on the primary detection endpoints of this synthetic campaign (FR 0.000 vs. 0.004; TTD_{ef} 0.000 vs. 0.101 s; F1 tied at 0.998), and at maximum suppression it reaches 93.6%, although at that operating point its detector fires on every episode (F1 = 0.866, the fire-always value). The exact zero in TTD_{ef} reflects pre-onset threshold activation: the Skip-RNN alert is already above θ before the onset in 21% of pre-onset frames, so detection collapses onto t_0 itself. This comparison exposed a limitation of the evaluation design – TTD_{ef} measures delay from the onset and does not penalize alerts raised before it – which is why the alert-occupancy metrics below were added. Under the frozen temporal-scale protocol of Section 4.12 (seeds 42–45), the frozen Skip-RNN also preserves the detection endpoints across the duration grid in that descriptive four-seed replay. We therefore do not rank the two designs on these endpoints; PreOcc and NormOcc are reported as *complementary signal-semantics metrics*, not as substitutes for the primary endpoints. On those measurements the emitted signals differ sharply: the Skip-RNN alert occupies 21% (matched) to 38% (maximum suppression) of the pre-onset frames of critical episodes and, at maximum suppression, 10.9% of the frames of normal episodes, whereas the AF-TOI signal remains silent on normal episodes (0.0%) and in 93–95% of the pre-onset phase, rising ≈18× at the transition (0.033 → 0.581 mean probability). The AF-TOI trajectory is thus an interpretable, temporally organized risk signal – the input the MHEG thresholds consume and the substance of the operational state ℓ_t – and it comes with the explicit runtime safety mechanisms (unconditional kinematic trigger, recalibratable thresholds, coverage constraint) that a learned gate does not provide. These are properties of the signal and of the runtime contract, not a claim of superiority on the detection endpoints above.

4.12. Generalization to Unseen Risk-Transition Durations

The stress campaign of Section 4.10 varies the signals available to the scheduler; it does not vary the temporal scale of the risk transition itself. To separate robustness across the predefined progressive and abrupt onset regimes from generalization across temporal scale, we compressed or expanded the original TERA-Gen risk-transition curves in time through a shape-preserving temporal warp. The deterministic onset t_0 remains a frame-precise instant; what is varied is the *risk-transition duration* d, the duration of the transition surrounding it. For this experiment, d is operationalized as the elapsed time from t_0 until the risk trajectory reaches 90% of its segment peak; it is therefore a measured quantity, distinct from the nominal ramp component of the generator described in Section 3.1, and is used only for the temporal-scale generalization analysis. The grid values are nominal targets of the warp; under discrete interpolation the realized first-crossing duration deviates from the nominal value by at most five frames (≈0.52 s at the slow end), so cells at the boundary of the training support partially straddle it. The warp preserves the normalized transition shape, t_0, and the episode context channels, and all model weights and scheduler thresholds of the twelve-seed campaign (seeds 42–53) are kept frozen: no re-training and no re-calibration are performed on any out-of-distribution condition. The grid spans nine

Table 20
Skip-RNN versus the frozen AF-TOI(+MHEG) models under a single harmonized protocol (main-campaign test partitions, seeds 42–45, $\theta = 0.10$, detection from the label onset; mean ± sd). "Matched" selects, per seed, the Skip-RNN cost weight whose suppression fraction is closest to MHEG's (a post-hoc diagnostic; suppression fractions, not FLOPs, are matched). PreOcc/NormOcc are the alert-occupancy metrics defined in the text (means over seeds).

System	FR↓	TTD_{ef} (s)↓	%Sup↑	F1↑	PreOcc↓	NormOcc↓
AF-TOI Fixed (frozen)	0.004 ± 0.008	0.086 ± 0.136	0	0.998 ± 0.004	0.068	**0.000**
AF-TOI+MHEG (frozen)	0.004 ± 0.008	0.101 ± 0.155	48.7 ± 7.3	0.998 ± 0.004	**0.051**	**0.000**
Skip-RNN (matched suppression)	0.000 ± 0.000	0.000 ± 0.000	50.9 ± 7.1	0.998 ± 0.004	0.211	0.001
Skip-RNN (maximum suppression)	0.000 ± 0.000	0.000 ± 0.000	93.6 ± 0.2	0.866 ± 0.000	0.375	0.109

durations from 0.21 to 5.21 s; the training distribution covers 0.42–1.04 s (abrupt) and 2.50–4.06 s (progressive), so the grid contains in-distribution anchors, an interpolation gap (1.46–2.29 s), and extrapolation extremes (0.21 and 4.38–5.21 s). At the native duration the warp reproduces the original trajectories exactly, and the evaluation machinery reproduces the original factorial campaign with zero error on the original test partitions of seeds 42–45. The TTD_{ef} magnitudes below are on this campaign's own scale – frame-level detection from the label onset on warped episodes – and are not comparable to the TTD_{ef} of Table 8, which uses the original episodes and the main-campaign pipeline (Table 23); the zero-error anchor concerns reproduction of the campaign artifacts, not a shared metric scale.

Figure 14a–b and Table 21 report the frozen operating points. AF-TOI maintains mean FR below 0.05 at every evaluated duration (maximum duration-wise mean 0.026), and TTD_{ef} stays within approximately 0.40–0.66 s. The Baseline crosses mean FR = 0.05 at both extrapolation extremes (mean FR 0.097 at 0.21 s and 0.085 at 5.21 s for the fixed policy), and its TTD_{ef} follows a U-shaped profile that reaches 2.12 s at the fast extreme. At the level of individual seed×duration cells, AF-TOI exceeds FR = 0.05 in 7 of 108 cells (all at $d \geq 3.23$ s, maximum 0.108), against 22 of 108 for the fixed Baseline (maximum 0.250) and 32 of 108 for Baseline+MHEG (maximum 0.333). Frame-level F1 remained seed-sensitive under the warp, a pattern consistent with checkpoint-level variability rather than with an effect of the warp itself. Accordingly, the result is restricted to aggregate coverage and effective delay and does not constitute a per-seed guarantee: AF-TOI shows aggregate temporal-scale robustness across the evaluated synthetic range of unseen risk-transition durations.

Because the two frozen gates suppress different fractions of frames, we added a post-hoc matched-budget diagnostic: for each seed×duration cell, the entropy threshold of Baseline+MHEG was adjusted by bisection – the budget-matching protocol of Section 4.10 – until its computation rate (fraction of executed window updates; both arms share the recompute semantics of Section 3.4) matched that of AF-TOI+MHEG on the same cell (all 108 cells matched within 1.9%, tolerance 2%; Figure 14d). Because the matching uses the evaluated condition itself, this is a diagnostic of allocation quality at equal cost, not a prospective deployment policy. At matched computation cost, no systematic FS@onset advantage remains: AF-TOI yields lower FS@onset in 47 of 108 cells, ties in 8, and higher in 53, and the two schedulers are statistically indistinguishable at eight of the nine durations (exact paired Wilcoxon, n=12, $p \geq 0.21$; p-values in this subsection are reported uncorrected for multiplicity across the nine durations). The single significant difference, at $d = 0.21$ s ($p = 0.027$), favors the matched Baseline – at the same duration where that configuration exhibits mean FR 0.111, i.e. outside the coverage constraint. The lower FS@onset observed under the frozen operating points is therefore largely attributable to the different global suppression budgets rather than to a uniform allocation advantage. FR and TTD_{ef}, in contrast, show uncorrected paired differences at matched cost at the extrapolation extremes (0.21 s: FR 0.006 vs. 0.111, $p = 0.008$; TTD_{ef} 0.66 vs. 2.29 s, $p = 0.003$; 5.21 s: $p = 0.042$ and $p = 0.007$) and at the 0.62-s abrupt in-distribution anchor ($p = 0.031$ and $p = 0.021$), with no significant differences in the interpolation gap and mid-range. Under Holm correction within each endpoint family, only the TTD_{ef} difference at 0.21 s remains significant ($p = 0.031$), so these per-duration matched-cost contrasts should be read as exploratory. The matched-budget diagnostic supports a temporal-scale coverage-and-timeliness advantage, not a universal improvement in near-onset suppression placement.

4.13. Embedded Deployment Viability

At runtime, the active infrastructure is limited to the causal LSTM target, the MHEG scheduler, constant thresholds, and the recurrent buffer (Table 22). AF-TOI+MHEG reaches 48.1% computation suppression (%Sup) and 49.8% effective cost reduction (ms/f, per-frame latency relative to Baseline Fixed), two metrics that jointly characterize the operational economy.

Estimates for dedicated embedded platforms come from scaling projections; physical chip validation is required before production. All embedded-viability figures in this subsection are analytical (parameter counts,

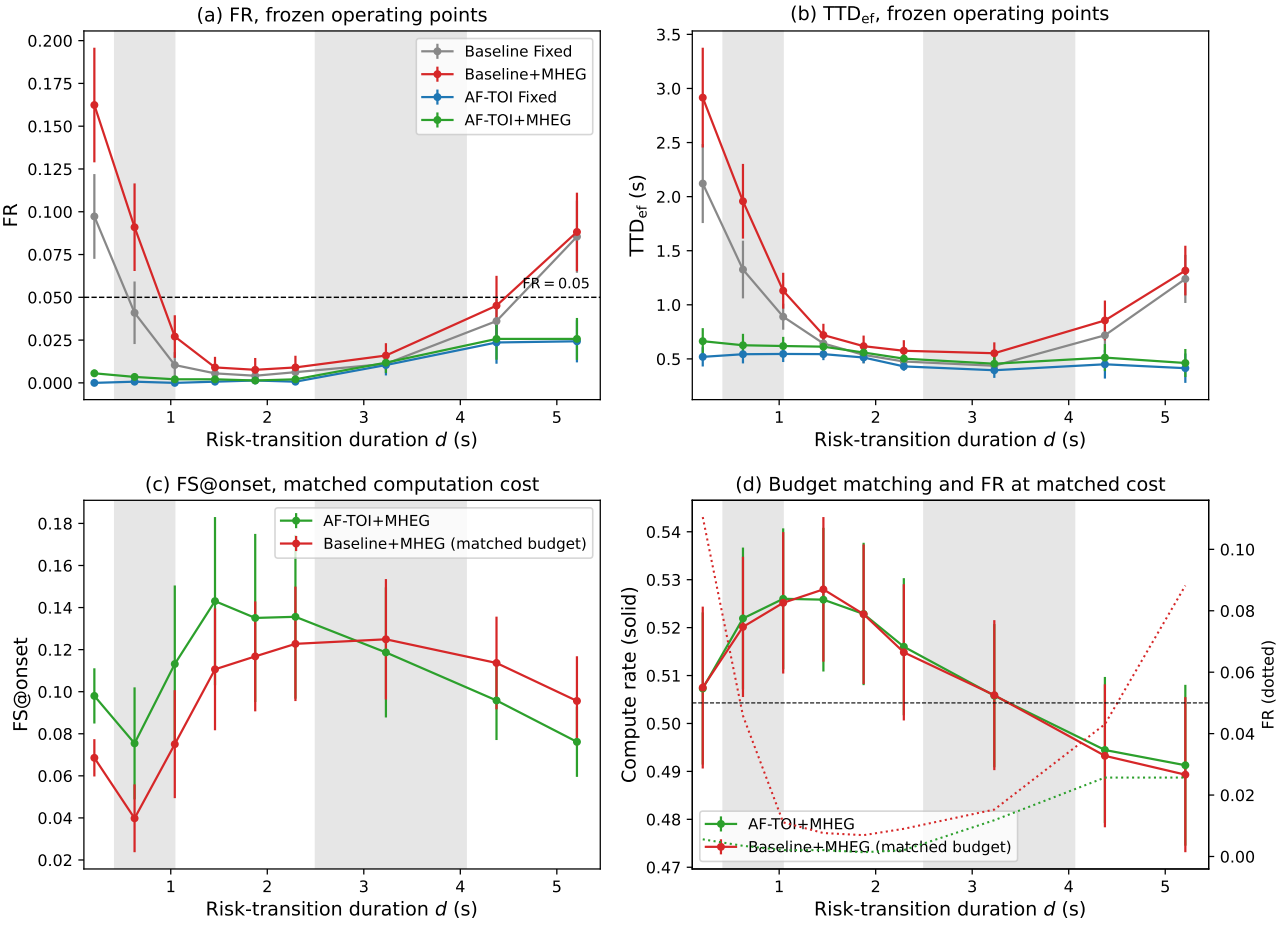


Figure 14: Temporal-scale generalization under shape-preserving risk-transition warps (twelve seeds; mean ± standard error). Shaded bands mark the temporal support represented in training (0.42–1.04 s abrupt, 2.50–4.06 s progressive); durations between and outside the bands are out-of-distribution. (a, b) FR and $\mathrm{TTD}_{\mathrm{ef}}$ at the frozen operating points. (c) Near-onset false suppression at matched computation cost; no systematic AF-TOI advantage is observed. (d) Computation rates after budget matching (solid, left axis) and the corresponding FR (dotted, right axis).

Table 21
Temporal-scale generalization across risk-transition durations d (mean ± sd, twelve seeds). FR and $\mathrm{TTD}_{\mathrm{ef}}$ (s) use the frozen operating points (fixed policy); FS@onset uses the post-hoc matched-budget diagnostic (Baseline+MHEG matched to the computation rate of AF-TOI+MHEG within 2%). Regime: ID = within training support; interp. = interpolation gap; extrap. = extrapolation.

		FR (frozen)		**$\mathrm{TTD}_{\mathrm{ef}}$** (s, frozen)		**FS@onset** (matched budget)	
d (s)	regime	Baseline	AF-TOI	Baseline	AF-TOI	Baseline+MHEG	AF-TOI+MHEG
0.21	extrap.	0.097±0.086	0.000±0.000	2.121±1.266	0.519±0.308	0.069±0.031	0.098±0.045
0.62	ID	0.041±0.063	0.001±0.002	1.326±0.924	0.543±0.294	0.040±0.056	0.075±0.092
1.04	ID	0.010±0.026	0.000±0.000	0.890±0.417	0.546±0.259	0.075±0.089	0.113±0.129
1.46	interp.	0.006±0.014	0.001±0.002	0.643±0.297	0.544±0.193	0.111±0.100	0.143±0.138
1.88	interp.	0.004±0.014	0.001±0.005	0.537±0.273	0.513±0.183	0.117±0.091	0.135±0.138
2.29	interp.	0.006±0.014	0.001±0.002	0.476±0.277	0.431±0.135	0.123±0.094	0.136±0.135
3.23	ID	0.011±0.012	0.010±0.021	0.436±0.269	0.395±0.244	0.125±0.099	0.119±0.107
4.38	extrap.	0.036±0.044	0.024±0.043	0.718±0.544	0.450±0.457	0.114±0.076	0.096±0.065
5.21	extrap.	0.085±0.073	0.024±0.043	1.239±0.770	0.414±0.470	0.096±0.074	0.076±0.058

Exact paired Wilcoxon (n=12) on FS@onset at matched budget: $p = 0.027$ at 0.21 s (favoring the matched Baseline, which exhibits mean FR 0.111 at this duration); $p \geq 0.21$ at all other durations.

Table 22
Embedded deployment viability indicators (unit-batch mode). FLOP figures use the 1 MAC = 2 FLOPs convention. ‡Evaluated windowed semantics (Section 3.4); the input-window figure is the explicit 96×31 FP32 window, an estimate rather than measured peak RAM.

Indicator	Baseline	AF-TOI	AF-TOI+MHEG
Parameters		≈ 58 K	
FP32 / INT8		≈ 228 KB / ≈ 57 KB	
Recurrent buffer		1024 bytes	
MACs/recurrent step		≈ 57.2 K (≈ 115 KFLOPs)	
MACs/update‡ (W=96)		≈ 5.49 M	
Input-window buffer‡		≈ 11.9 KB	
Latency (ms/f)	0.677	0.665	**0.340**
Computation suppression	0%	0%	48.1%

memory footprints, FLOP counts) or measured on a reference GPU; none derives from execution on physical automotive hardware.

Experimental evaluation synthesis Taken together, the six movements converge on a consistent picture within this controlled synthetic, single-node setting: temporal differentiation is what enables coverage-preserving selective inference, with driver monitoring as the concrete case. Under the budget-matched diagnostics of Sections 4.10 and 4.12, the advantage concentrates on coverage and detection timeliness under the frozen operating policy, rather than on where computation lands at strictly equal cost. AF-TOI induces the differentiation that makes MHEG's criterion discriminative, so computation goes to emergence windows and is withheld through redundant ones; the gain traces to a reorganized failure distribution rather than a lower mean, it holds across seeds, and – descriptively, under a second evaluation protocol – it carries over to intermediate-risk episodes never seen in training. All of this runs within ≈58 K parameters, a 1,024-byte recurrent state, and – in the evaluated windowed semantics – an ≈11.9 KB input-window buffer (Section 3.4); the parameter and state footprints are within reach of automotive-grade microcontrollers and low-power accelerators, while the per-update compute depends on which of the two execution semantics is deployed.

Artifact availability Source code and TERA-Gen synthetic datasets are publicly available as described in the Data Availability statement.

Table 23
Experimental protocol map. "Frozen" means weights and thresholds fixed before the experiment; the replication protocol is the replay pipeline of Sections 4.10 and 4.11, distinct from the main-campaign pipeline. Values are comparable within a row only.

Experiment (location)	Seeds	Models	Protocol, decision rule, primary metrics	Budget / ECE
Main factorial (§4.4–4.6; Tabs. 8–10, 12)	42–53	trained per seed, frozen TERA pipeline	F1 episode rule: mean of last k=6 frames vs. validation-calibrated threshold; FR and TTD from the stable-detection rule (θ=0.10, m=3, from onset); $\mathrm{TTD}_{\mathrm{ef}}$, %Sup	— / $\mathrm{ECE}_{\mathrm{pipe}}$
Borderline, Movement 6 (Tab. 13, Fig. 11)	42	frozen factorial models	held-out borderline recipes; $\bar{p}_{\max}$ descriptor and activation fractions at θ=0.10	—
Post-hoc calibration (§4.8, Tab. 14)	42–45	frozen Baseline + fitted calibrators	pipeline trajectories; MHEG thresholds recalibrated per variant; FR, $\mathrm{TTD}_{\mathrm{ef}}$, %Sup	— / $\mathrm{ECE}_{\mathrm{cal}}$
Teacher ablation (§4.9, Tab. 15)	42–45	retrained, simplified protocol	simplified training (60 epochs, no scheduler); not comparable to the main study	—
Mechanism ablation (§4.6, Tab. 11)	42–53	all five arms retrained under the replication protocol	fixed policy and MHEG contrasts; F1, FR, $\mathrm{TTD}_{\mathrm{ef}}$, %Sup	— / $\mathrm{ECE}_{\mathrm{v29}}$
Stress campaign (§4.10, Tabs. 16–18)	42–45	frozen replication-protocol trajectories	replay over nine scenarios; FS@onset, coverage, miss	compute-rate matched ±2%
Gating baselines (§4.11, Tab. 19)	42–45	frozen replication-protocol trajectories	replay; oracle thresholds on test; detection from label onset, θ=0.10, m=3	matched suppression
Skip-RNN (§4.11, Tab. 20)	42–45	Skip-RNN trained end-to-end; AF-TOI(+MHEG) frozen	harmonized protocol; label onset, θ=0.10, m=3; PreOcc, NormOcc	matched suppression
Temporal warp (§4.12, Tab. 21, Fig. 14)	42–53	frozen twelve-seed models and thresholds	warped TERA-Gen episodes; label-onset detection; FR, $\mathrm{TTD}_{\mathrm{ef}}$, FS@onset	bisection-matched compute
Deployment (§4.13, Tab. 22)	—	deployed causal model	analytical counts + reference-GPU latency	—

5. Discussion

Temporal organization as the enabling condition for edge resource allocation The most informative result is the negative control, more than the full configuration. Baseline+MHEG buys computational savings, but not a safety-qualified efficiency gain: the miss rate is not improved and the effective detection delay worsens. This locates AF-TOI's primary effect. The mechanism does not work by raising classification scores; it creates the conditions under which the evaluated MHEG policy achieves coverage-preserving selective computation, under the controlled synthetic conditions studied here. Without differentiated dynamics, efficiency and coverage do not merely fail to improve together; they compete, because the scheduler trims apparent per-frame cost while the effective delay climbs. Once the trajectories are organized, that competition relaxes (Movements 4.2 and 4.3): H_t turns into a variable the scheduler can discriminate on, concentrating computation at high-instability timesteps and reusing buffers through the stable ones. The premise behind confidence- and uncertainty-driven selective inference (that low uncertainty marks content safe to skip) holds in activity recognition; in our controlled causal replay it fails for unorganized trajectories, and the negative control is where that failure becomes measurable.

The Skip-RNN comparison of Section 4.11 adds the complementary nuance: a learned gate can still reach the in-distribution detection endpoints, at the pre-onset alert occupancy reported there. In summary, the effectiveness of the runtime allocator depends on a property the model must already possess, namely temporal differentiation in its trajectories.

Two in-distribution caveats bound this reading: isotonic recalibration combined with MHEG matches or exceeds AF-TOI+MHEG on the replay endpoints of Section 4.8, and a periodic-refresh confidence gate suffices for the replay endpoints of Section 4.11; a published learned gate, Skip-RNN, likewise matched or exceeded AF-TOI+MHEG on the conventional detection endpoints at matched suppression (Section 4.11). We therefore do not claim superiority over learned adaptive computation on detection performance; its substantially higher pre-onset alert occupancy is what exposed a limitation of onset-delay metrics and motivated the signal-semantics analysis. What none of these alternatives provides is the interpretable, onset-anchored signal semantics. Their behavior under the controlled shifts of Sections 4.10 and 4.12 is unequally characterized: Skip-RNN was run under the temporal-scale warp (descriptively, preserving its detection endpoints), while the isotonic and periodic-refresh alternatives were not tested under shift. A limitation of $\mathrm{TTD}_{\mathrm{ef}}$ itself follows from the Skip-RNN result: the metric captures missed-event and post-onset delay costs (Eq. 15) but does not penalize persistent pre-onset activation, so it should not be read as a complete operational utility metric. It remains valid for the purpose it was defined for – arbitrating post-onset timeliness under an explicit miss penalty – while PreOcc and NormOcc expose the complementary pre-onset dimension; we deliberately do not aggregate these into a single composite score, leaving the joint reporting of the three quantities as the stated protocol.

AF-TOI as enabler of trustworthy selective computation A model with 95% classification accuracy and an unsafe scheduler is not equivalent to one whose scheduler allocates correctly at emergence windows. In this controlled synthetic setting, TERA provides evidence that scheduler quality is at least as important as classification quality, and that protocols optimizing only accuracy or calibration do not guarantee the trajectory structure scheduling reliability depends on, an implication for edge-native ITS design.

Probabilistic calibration and temporal organization as independent edge design properties The post-hoc calibration control carries a direct design implication. Temperature, Platt, and isotonic scaling improve aggregate alignment without touching weights or internal trajectories, and the $H(t)$ morphology around onset stays within the Baseline range for all three: none of the evaluated calibrators induced temporal differentiation in the measured trajectories. Where PS and IC reach lower $\mathrm{TTD}_{\mathrm{ef}}$ under MHEG, the unchanged entropy-trajectory morphology attributes the gain to threshold recalibration on the redistributed scale rather than to organizational restructuring. In the configurations measured here, temporal induction and probability-scale adjustment produced distinct operating points (Figure 13); whether that separation persists under distribution shift was not tested. Marginal calibration and temporally conditioned operational reliability are empirically dissociable under this protocol, with distinct mechanistic origins.

Effective coverage versus gross delay The Baseline shows lower gross delay on detected episodes, but that metric omits misses. For safety-critical edge systems, coverage across all episodes matters more than mean latency on successes, and any metric excluding misses reads optimistically.

Complementary role of the kinematic component When pre-onset windows fall below 0.6 s, temporal differentiation may not appear in entropy fast enough. MHEG's kinematic component covers this by triggering allocation unconditionally when instantaneous physical variation signals a sudden state change. The disjunction between components handles two emergency regimes with distinct dynamics without depending exclusively on entropy, which matters for robustness under abrupt onset.

Implications for edge AI deployment and vehicular edge integration Most deployment properties follow from the offline/online separation. The computationally intensive stages (TERA-Gen, the BiLSTM reference, AF-TOI, and threshold calibration) remain in the development environment; the device runs only the causal LSTM, the calibrated thresholds, and the recurrent buffer, with no reference model to execute and no online retraining. MHEG's own overhead per timestep is $\mathcal{O}(K)$ for the kinematic gate plus $\mathcal{O}(1)$ for the entropy and level thresholds, negligible beside the recurrent update it gates. At ≈58 K parameters, a 1,024-byte recurrent state, and ≈57.2 K MACs (≈115 KFLOPs) per recurrent step – W such steps per executed update in the evaluated windowed semantics, plus an ≈11.9 KB input-window buffer (Section 3.4), the model's parameter and state footprints sit within the envelope of current automotive-grade microcontrollers and accelerators, and because the only artifacts that move to the device are calibrated weights and thresholds, it is compatible with OTA updates and needs no connectivity at inference time. The ℓ_t signal, whose level is obtained from p_t by an $O(1)$ threshold mapping (Eq. 13), is the natural point of contact with higher-level VEC and MEC coordination. That contact, however, is an implication we develop in Section 6; it is not evaluated here.

Future directions Three extensions follow. First, the MHEG scheduling signal could feed future VEC and MEC offloading policies (Section 6), triggering remote inference when local entropy exceeds a coordination threshold while suppressing redundant timesteps and reallocating the saved energy. Second, online adaptation of MHEG thresholds under distributional shift is the principal step toward production. Methodological precedents from another sensing domain provide useful design references for this direction. In device-free Wi-Fi localization, online learning has been used to adapt models across domains without full retraining [52], and online spatiotemporal modeling has kept lightweight models robust in nonstationary environments [53]. Whether such incremental schemes transfer to onset-anchored selective computation remains open, since any adaptation of the MHEG gate must stay causal and must continue to honor the coverage constraint $\mathrm{FR} \leq 0.05$. Third, adversarial robustness under sensor faults, malicious perturbations, and modality degradation is a natural safety extension.

Scope and applicability The campaign is scoped to a single-node deployment model running continuous causal inference (executed on the reference GPU of Section 4.13, not on embedded hardware), which isolates the temporal-organization effect from network-level confounders. No networked coordination is evaluated. The mechanism could be reused in future distributed deployments; Section 6 discusses how the MHEG entropy gate could inform VEC and MEC offloading, with empirical validation under real network conditions left as future work. Robustness against adversarial attacks, sensor faults, and cybersecurity threats is out of scope and noted in Section 5.

Methodological rationale for synthetic evaluation The systematic use of TERA-Gen is deliberate. Controlled evaluation of temporal organization needs three properties naturalistic datasets cannot provide together at the required granularity: (i) *deterministic onset annotation*, removing the ≈0.31 s variability that would make scheduler differences indistinguishable at 9.6 fps; (ii) *causal isolation*, preventing correlated naturalistic confounders from masking the effect of trajectory structure on selectivity; and (iii) *frame-precise attribution*, enabling exact quantification of when the scheduler allocates relative to onset. TERA-Gen anchors synthetic trajectories to SHRP2 odds ratios [15], preserving statistical realism while achieving annotation precision naturalistic collection cannot guarantee. It does not replace naturalistic evaluation; it makes the hypothesis testable at the required granularity. Generalization to naturalistic data is the primary direction for future validation (Limitation 1).

Threats to validity and deployment limitations Four limitations bound the claims. First, evaluation relies on TERA-Gen synthetic data, designed to isolate the causal relationship between temporal organization and trustworthy selective computation under controlled onset; confirmation on naturalistic datasets such as DMD [2] and DADA [16] remains necessary for external validity. Second, MHEG thresholds are calibrated on synthetic data and may need re-calibration under real-world shift from driver variability, illumination, and sensor noise.

Third, the factorial analysis uses twelve seeds (42–53) under a prespecified protocol: the eight seeds beyond the initial four (42–45) were fixed before inspecting their outcomes, and all twelve are reported; with $n = 12$ the exact paired Wilcoxon test reaches conventional significance on F1, ECE, and $\mathrm{TTD}_{\mathrm{ef}}$ (Section 4.4). The remaining statistical caveat is narrower: FR in the fixed contrast is at a ceiling (most seeds have zero misses in both configurations), and inter-seed replication under a single synthetic generator quantifies sensitivity to initialization, not external validity – significance here does not substitute for naturalistic validation. The mechanistic anchor remains the negative control, where MHEG on unorganized trajectories worsens $\mathrm{TTD}_{\mathrm{ef}}$ while the same scheduler on organized trajectories preserves coverage and cuts computation.

Fourth, deployment feasibility is assessed through profiling and analytical estimation rather than execution on physical automotive hardware, so latency, energy, thermal behavior, and sensor-noise robustness remain to be characterized on STM32H7-, Cortex-M55-, and SA8155P-class platforms.

A related caveat is that a controlled environment could, in principle, favor an onset-oriented training strategy. Three design elements mitigate, without eliminating, this risk: episodes mix progressive and abrupt regimes in fixed proportion (Section 3.1); Movement 6 evaluates recipe families never seen in training; and the stress campaign of Section 4.10 spans nine onset/noise scenarios, including degraded entropy, lagged entropy, missing kinematic channels, and false onsets. Temporal-scale generalization is evaluated explicitly under shape-preserving synthetic warps with frozen models and thresholds (Section 4.12): AF-TOI maintains the duration-wise mean FR within the operational bound across risk-transition durations of 0.21–5.21 s (with seed-level exceedances in 7 of 108 cells), but this controlled temporal OOD test does not establish generalization to naturalistic distribution shift, to transition-shape families beyond those generated by TERA-Gen, or to unseen sensing statistics. Generalization across risk-trajectory families beyond these, and under real distribution shift, remains to be established on naturalistic data.

Planned real-data validation As the primary next step we will test whether the temporal organization induced on TERA-Gen transfers to real driving video, using the DADA2000 dataset [16], which provides frame-level accident annotations – an abnormal-onset frame, an accident frame, and an abnormal-end frame – together with annotated contextual attributes (weather, lighting, scene, and road geometry). The abnormal-onset frame serves as a real, frame-precise onset t_0, and the interval up to the accident frame is the emergence window in which detection must occur; this is the property naturalistic driver datasets usually lack and that TERA-Gen was built to supply synthetically. The scene and contextual attributes map onto the visual and contextual channels of our feature schema.

Because DADA2000 is scene-facing and carries no ego-vehicle kinematics, the kinematic trigger of MHEG has no native input; we will therefore evaluate the entropic gate both on its own and with an optical-flow-based kinematic proxy, an ablation that isolates the contribution of the induced organization from that of the kinematic trigger. Evaluation will use clip-level splits stratified by scene and accident type, with clips from the same source kept on the same side to prevent leakage, and MHEG thresholds re-calibrated per split on real validation data and never transferred across domains.

We will compare three transfer regimes – zero-shot, with the organization-bearing recurrent layers frozen and only the input projection adapted; light fine-tuning, with the recurrent layers frozen and the input and output layers adapted; and full re-induction of AF-TOI on the real training split – against the same negative control used here (Baseline+MHEG).

Success is defined not by matching the synthetic magnitudes, which are on a different scale, but by reproducing the qualitative relationships established on TERA-Gen: the negative control failing to improve

coverage, and AF-TOI+MHEG preserving coverage while suppressing computation. On the real data we will report $\mathrm{TTD}_{\mathrm{ef}}$, FR, %Sup, and near-onset false suppression (FS@onset, Section 4.10) with the dataset's own episode length and sample size, alongside the accident-anticipation metrics standard for that dataset. No real-data results are reported in this paper; this paragraph states the planned protocol only, and the entropy-lag tolerance measured in Section 4.10 sets the expectation that the scheduler should absorb the timing imprecision of real annotations up to a bounded lag.

6. Architectural Implications for Vehicular Edge Computing and Future Offloading

This section offers an architectural interpretation, not an evaluated distributed system. The campaign in Section 4 ran in a single-node configuration under a fixed budget (reference-GPU execution; Section 4.13): no networked coordination, offloading, communication latency, or bandwidth analysis was evaluated. The validated contribution remains single-node embedded selective computation. Here we explain only how the node-level operational state signal ℓ_t, validated in that setting, could inform future VEC and MEC architectures.

6.1. TERA as a Source of Edge Operational State Signals

The MHEG runtime output is a discrete signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, emitted once per timestep, with the level obtained from p_t by an $O(1)$ threshold mapping (Algorithm 1, Eq. 13). It is compact, a two-bit integer that can be stored, transmitted, or logged without reference to LSTM weights, recurrent state, or sensor frames, and it is interpretable through the epidemiological calibration of TERA-Gen (Table 3). When MHEG suppresses the update, ℓ_t comes from the buffered p_{t-1} with no inference at all.

What sets ℓ_t apart from a thresholded classifier output is its temporal grounding. Under AF-TOI, ℓ_t reflects phase-conditioned trajectory structure rather than instantaneous confidence, and the negative control of Section 4.4 shows a Baseline scheduler without this structure degrades $\mathrm{TTD}_{\mathrm{ef}}$ by 13.9%. This work's contribution is producing ℓ_t with experimentally supported temporal grounding; how it is consumed externally is a downstream question for future work.

6.2. Candidate Integration Paths for VEC and MEC Offloading

Resource allocation at the embedded tier The most immediate implication is within the on-board unit. Under AF-TOI+MHEG, 48.1% of frames are suppressed without violating FR ≤ 0.05, freeing cycles that could support other on-board functions. Whether such sharing is feasible depends on the hardware platform and RTOS scheduler, which this work does not evaluate.

Compatibility with VEC and MEC offloading In VEC and MEC architectures, the on-board unit decides which tasks to keep local and which to offload to roadside or cloud infrastructure. The signal ℓ_t could serve as one urgency input among others, with persistent Alert or Critical indicating time-critical conditions and Normal or Attention indicating deferrable ones. Such a policy could transmit only the compact symbol ℓ_t plus a summary feature window rather than raw frames, consistent with bandwidth-constrained partial offloading [54]. The activation fractions in Movement 6 (Section 4.7) are episode-level severity indicators, not measured communication rates; translating them into frame-level communication load requires network-level validation not performed here.

From scheduling signal to coordination primitive. In the borderline setting of Movement 6, activation fraction was more informative than instantaneous mean probability for summarizing the response to intermediate-risk episodes. A future policy could escalate on the persistence of elevated levels over a short window rather than on single readings, expressible as a rolling escalation statistic,

$$\Phi_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} \mathbf{1}[\ell_{t-i} \geq 2]\,, \tag{16}$$

the fraction of the last w timesteps at Alert or Critical. $\Phi_t(w)$ adds only a bounded moving sum over a symbol the system already emits, at a memory cost of w two-bit symbols, negligible next to the 1,024-byte buffer. It is not evaluated here but offered as a candidate bridge between the validated single-node mechanism and a future multi-tier policy, consistent with distributed early-exit designs [55] and VEC offloading frameworks [54, 56]; a future system could combine it with channel quality, queue length, and latency budget.

In summary, the single-node mechanism validated here is shaped, without modification, as a candidate coordination primitive for future VEC and MEC deployments, and the finding that activation frequency summarized escalating risk more stably than instantaneous mean probability could inform how a future escalation policy is designed. Empirical validation under real network conditions remains future work.

7. Conclusion

We set out to test whether selective computation can be made safe in strictly causal vehicular edge inference, and to identify what the safety actually depends on. Our hypothesis was that it depends less on the scheduler's policy than on the temporal organization of

the model's predictive trajectories – the property that makes uncertainty informative in the seconds around a critical transition. Within the controlled setting studied here, the evidence supports that reading, and the negative control is what makes it credible: with the scheduler applied to unorganized trajectories, $\mathrm{TTD}_{\mathrm{ef}}$ worsened by 13.9% (twelve seeds, $p = 0.001$) and coverage did not improve, so activating a scheduler is not, by itself, enough. Given AF-TOI-organized trajectories, the same MHEG scheduler reduced per-frame inference cost by 49.8% relative to Baseline Fixed (0.340 vs. 0.677 ms/frame), with $\mathrm{TTD}_{\mathrm{ef}}$ of 0.099 s against 0.570 s under Baseline+MHEG.

The post-hoc calibration control addressed the competing explanation: under this protocol, probabilistic rescaling improved ECE without reorganizing entropy in time, so marginal calibration and temporally conditioned reliability are empirically dissociable under this protocol. The budget-matched diagnostics bound the claim: at strictly equal cost, the advantage is in coverage and detection timeliness under the frozen operating policy, not in a uniform allocation advantage (Section 4.12).

The design lesson is not specific to driver monitoring. A training protocol that optimizes only classification accuracy or aggregate calibration can still leave a model unsafe to schedule on, because neither objective constrains the internal trajectory structure a causal scheduler reads. On this evidence, temporal organization is worth treating as a design property in its own right for resource-constrained edge AI, rather than as an accidental byproduct of training.

The pipeline ends in the operational-risk signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, whose level is obtained from the emitted probability by an $O(1)$ threshold mapping, at the end of the chain *Temporal Organization* $\rightarrow$ *Entropy Stratification* $\rightarrow$ *MHEG*. Because it is compact, real-time, and interpretable at the level of the trajectory (Section 6.1), it is a plausible primitive for distributed and vehicular edge architectures to consume without touching the onboard loop – though we present that use as a design path (Section 6), not a validated result.

The scope of the present validation should be stated explicitly. TERA is validated as a temporal-organization and scheduling substrate for a *single* embedded node, under strictly causal inference, on controlled synthetic data with a deterministic onset; the deployed model fits in ≈58 K parameters and a 1,024-byte recurrent state, with the evaluated windowed semantics additionally buffering the ≈11.9 KB input window (Section 3.4), within reach of automotive-grade platforms. Controlled temporal-scale OOD tests further show aggregate robustness of coverage and effective delay under unseen synthetic risk-transition durations (Section 4.12). It is not validated as a standalone detection system, and no networked coordination, offloading, or multi-node behavior was evaluated. The VEC and MEC directions of Section 6 are implications of the ℓ_t signal, to be tested rather than claimed. Accordingly, the immediate priorities are external validity on naturalistic data, robustness under distributional shift, sensor faults, and adversarial perturbation, characterization on physical automotive hardware, and an empirical multi-node study of the offloading policies sketched here – each building on the single-node, entropy-gated mechanism this paper characterizes.

Authors' Contribution Statement (CRediT)

Ângela Brígida Albarello: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing (original draft, review and editing). **Edna Dias Canedo**: Academic Supervision, Conceptualization, Writing (review and editing). **Vinícius P. Gonçalves**: Writing (review and editing). **Leandro A. Villas**: Supervision, Writing (review and editing). **Rodolfo I. Meneguette**: Writing (review and editing). **Geraldo P. Rocha Filho**: General Supervision, Writing (review and editing).

Declaration of Competing Interests

The authors declare no competing financial interests or personal relationships that could have influenced the reported results.

Data Availability

The source code of the frozen TERA pipeline, the synthetic data generator, the configuration files, the twelve-seed campaign outputs, and the evaluation and figure scripts are publicly available at `https://github.com/angelabalbarello/tera-adhoc-artifact`. The synthetic datasets are fully reproducible from the released generator and configuration hashes.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the

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